Chapter 7 Sampling Distributions

- In Chapter 1 we **introduced** random sampling.
- In this chapter we explain what a random sample is and how to select a random sample.
- In addition, we discuss two probability distributions that are related to random sampling.
- To understand these distributions, note that if we select a random sample, then we use the sample mean as the point estimate of the population mean and the sample proportion as the point estimate of the population proportion.

Why we need to study the Sampling Distribution ?

Example: The average height of China's year one university students

- There are about 3,000,000 year one university students in China. It is not possible (too costly) to survey every student.
- Randomly select 500 students from the 3,000,000 and get their heights.
- Calculate the average height of these 500 students and get, for example, a result of 172.5 cm.
- This 172.5 cm is a Sample Mean.

Now, the crucial question arises:

- Can this 172.5 cm represent the true average height of the entire 3,000,000 students? Is it too high? If a different group of 500 students were picked, would the result be very different?

To answer this question, we need to introduce the "sampling distribution of the sample mean."

- In Chapter 3, we have mentioned below formula for the variance of sample of size *n*.

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n-1}$$

In this Chapter, we will answer why we divide by $n - 1$ but not n.

Section 7.1 Random Sampling

Section 7.2. The Sampling Distribution of the Sample Mean.

Section 7.3 The Sampling Distribution of the Sample Proportion.

Section 7.1

Random Sampling

Selecting a random sample from a population is one of the best ways to ensure that the information contained in the sample reflects what is true about the population. To illustrate the idea of a random sample, consider the cell phone case.

Random (or approximately random) sampling—as well as the more advanced kinds of sampling discussed in optional Section 7.4—are types of probability sampling. In general, probability sampling is sampling where we know the chance (or probability) that each element in the population will be included in the sample. If we employ probability sampling, the sample obtained can be used to make valid statistical inferences about the sampled population.

Example 7.1 The Cell Phone Case: Reducing Cellular Phone Costs

In order to select a random sample of 100 employees from the population of 2,136 employees, the bank will make a numbered list of the 2,136 employees. The bank can then use a random number table, such as Table 7.1(a) (next page), to select the random sample.

Arbitrarily select any set of four digits in the table (circled).

Do so until 100 employees have been found.

TABLE 7.1 Random Numbers

(a) A portion of a random number table

33276	85590	79936	56865	05859	90106	78188
03427	90511	69445	18663	72695	52180	90322
92737	27156	33488	36320	17617	30015	74952
85689	20285	52267	67689	93394	01511	89868
08178	74461	13916	47564	81056	97735	90707
51259	63990	16308	60756	92144	49442	40719
60268	44919	19885	55322	44819	01188	55157
94904	01915	04146	18594	29852	71585	64951
58586	17752	14513	83149	98736	23495	35749
09998	19509	06691	76988	13602	51851	58104
14346	61666	30168	90229	04734	59193	32812
74103	15227	25306	76468	26384	58151	44592
24200	64161	38005	94342	28728	35806	22851
87308	07684	00256	45834	15398	46557	18510
07351	86679	92420	60952	61280	50001	94953

While using a random number table is one way to select a random sample, this approach has a disadvantage that is illustrated by the current situation.

Obtaining 100 different, four-digit random numbers between 0001 and 2136 will require ignoring a large number of random numbers in the random number table.

So, using “Random number table” is very time-consuming, in many cases.

A good alternative is to use a computer software package, which can generate random numbers that are between whatever values we specify. For example, Table 7.1(b) gives the MINITAB output of 100 different, four-digit random numbers that are between 0001 and 2136 (note that the “leading 0’s” are not included in these four-digit numbers)

TABLE 7.1 Random Numbers

(b) MINITAB output of 100 different, four-digit random numbers between 1 and 2136

705	1131	169	1703	1709	609
1990	766	1286	1977	222	43
1007	1902	1209	2091	1742	1152
111	69	2049	1448	659	338
1732	1650	7	388	613	1477
838	272	1227	154	18	320
1053	1466	2087	265	2107	1992
582	1787	2098	1581	397	1099
757	1699	567	1255	1959	407
354	1567	1533	1097	1299	277
663	40	585	1486	1021	532
1629	182	372	1144	1569	1981
1332	1500	743	1262	1759	955
1832	378	728	1102	667	1885
514	1128	1046	116	1160	1333
831	2036	918	1535	660	
928	1257	1468	503	468	

Exercises for Section 7.1

Companies:

1 Altria Group
2 PepsiCo
3 Coca-Cola
4 Archer Daniels
5 Anheuser-Bush
6 General Mills
7 Sara Lee
8 Coca-Cola Enterprises
9 Reynolds American
10 Kellogg
11 ConAgra Foods
12 HJ Heinz
13 Campbell Soup
14 Pepsi Bottling Group
15 Tyson Foods

7.3 In the page margin, we list 15 companies that have historically performed well in the food, drink, and tobacco industries. Consider the random numbers given in the random number table of Table 7.1(a) (slide 9, and also the textbook). Starting in the upper left corner of Table 7.1(a) and moving down the two leftmost columns, we see that the first three two-digit numbers obtained are: 33, 03, and 92. Starting with these three random numbers, and moving down the two leftmost columns of Table 7.1(a) to find more two-digit random numbers, use Table 7.1(a) to randomly select five of these companies to be interviewed in detail about their business strategies.

Exercises for Section 7.1

7.4 The bank customer waiting time case (Optional)

Recall that when the bank manager's new teller system is operating consistently over time, the manager decides to record the waiting times of a sample of 100 customers that need teller service during peak business hours. For each of 100 peak business hours, the first customer that starts waiting for service at or after a randomly selected time during the hour will be chosen. Consider the peak business hours from 2:00 P.M. to 2:59 P.M., from 3:00 P.M. to 3:59 P.M., from 4:00 P.M. to 4:59 P.M., and from 5:00 P.M. to 5:59 P.M. on a particular day. Also, assume that a computer software system generates the following four random numbers between 00 and 59: 32, 00, 18, and 47. This implies that the randomly selected times during the first three peak business hours are 2:32 P.M., 3:00 P.M., and 4:18 P.M. What is the randomly selected time during the fourth peak business hour?

Answer: 5:47 P.M.

Section 7.2

Sampling Distribution of the Sample Mean

The sampling distribution of the sample mean $\bar{x}$ is the probability distribution of the population of the sample means obtainable from all possible samples of size n from a population of size N.

In other words, the probability distribution of $\bar{x}_1, \bar{x}_2, \ldots,$ for all samples.

Recall: Combination

- The combination is a way of selecting items from a collection, such that the order of selection does not matter.
- For example, given three elements, say A, B and C, there are three combinations of two that can be drawn from this set: AB, AC, BC.
- Combination Formula: $C_n^k = \binom{n}{k} = \frac{n!}{k!(n-k)!}$
- For example, $n = 3$, $k = 2$,

$$C_3^2 = \binom{3}{2} = \frac{3!}{2!\,(3-2)!} = 3$$

- Result: There are $C_6^2 = \binom{6}{2} = \frac{6!}{2!(6-2)!} = 15$ possible samples of size $n = 2$
- Calculate the sample mean of each and every sample.
- For example, if we choose the two stocks with returns 10% and 20%, then the sample mean is 15%.

7.2.1 Sample Mean

- Let there be a population of units of size N.
- Consider all its samples of a fixed size n ($n<N$).
- For all possible samples of size n, we obtain a population of sample means. That is, $\bar{x}$ ***is a random variable*** which may have all these means as its values.
- Before we draw the sample, the sample mean $\bar{x}$ is a random variable.
- We consider the probability distribution of the random variable $\bar{x}$, i.e., the probability distribution for the population of sample means.

Example: The law firm of Hoya and Associates has five partners. At their weekly partners meeting each reported the number of hours they billed clients for their services last week.

Partner	Hours
Dunn	22
Hardy	26
Kiers	30
Malory	26
Tillman	22

If two partners are selected randomly, how many different samples are possible?

Partners	Total	Mean
1,2	48	24
1,3	52	26
1,4	48	24
1,5	44	22
2,3	56	28
2,4	52	26
2,5	48	24
3,4	56	28
3,5	52	26
4,5	48	24

A total of 10 different samples

As a sampling distribution

Sample Mean	Frequency	Relative Frequency probability
22	1	1/10
24	4	4/10
26	3	3/10
28	2	2/10

Different samples of the same size from the same population will yield different sample means.

- Here the population is the 5 partners.
- Here the size of the samples is 2.
- We get a population of sample means, 22, 26, 24, 22, 28, 26, 24, 28, 26, and 24.
- The size of the population of sample means is 10.

In this example,

- the population mean is $\mu = \frac{22+26+30+26+22}{5} = 25.2$
- the mean of all sample means is

$$\mu_{\bar{x}} = \frac{24 + 26 + 24 + 22 + 28 + 26 + 24 + 28 + 26 + 24}{10} = 25.2$$

- It can be seen that $\mu_{\bar{x}} = \mu$

7.2.2 Sampling Distribution

The sampling distribution of the sample mean is the probability distribution of the population of the sample means obtainable from all possible samples of size n from a population of size N.

The sampling distribution of the sample mean:

- It is a probability distribution.
- It is the distribution of the population of the sample means.
- The sample means are obtained from all possible samples.
- These samples have same size n and are from a population of size N.

Example 7.2: The Car Mileage Case: Estimating mean mileage

- Last year the automaker made six preproduction cars of a new model.
- Two of these six cars were randomly selected for testing.
- When this was done, the cars obtained mileages of 30 mpg and 32 mpg. The mean of this sample of mileages is

$$\bar{x} = \frac{30 + 32}{2} = 31 \text{ mpg}$$

This sample mean is the point estimate of the mean mileage μ for the population of six preproduction cars and is the preliminary mileage estimate for the new model.

- Late, the automaker decided to further study the new model by subjecting the rest four cars. When the mileage test was performed, the four cars obtained mileages of 29 mpg, 31 mpg, 33 mpg, and 34 mpg.
- Thus, the mileages obtained by the six preproduction cars were 29 mpg, 30 mpg, 31 mpg, 32 mpg, 33 mpg, and 34 mpg. The probability distribution of this population of six individual car mileages is given in Table 7.2 and graphed in Figure 7.1(a). The mean of the population of car mileages is

$$\mu = \frac{29 + 30 + 31 + 32 + 33 + 34}{2} = 31.5 \text{ mpg}$$

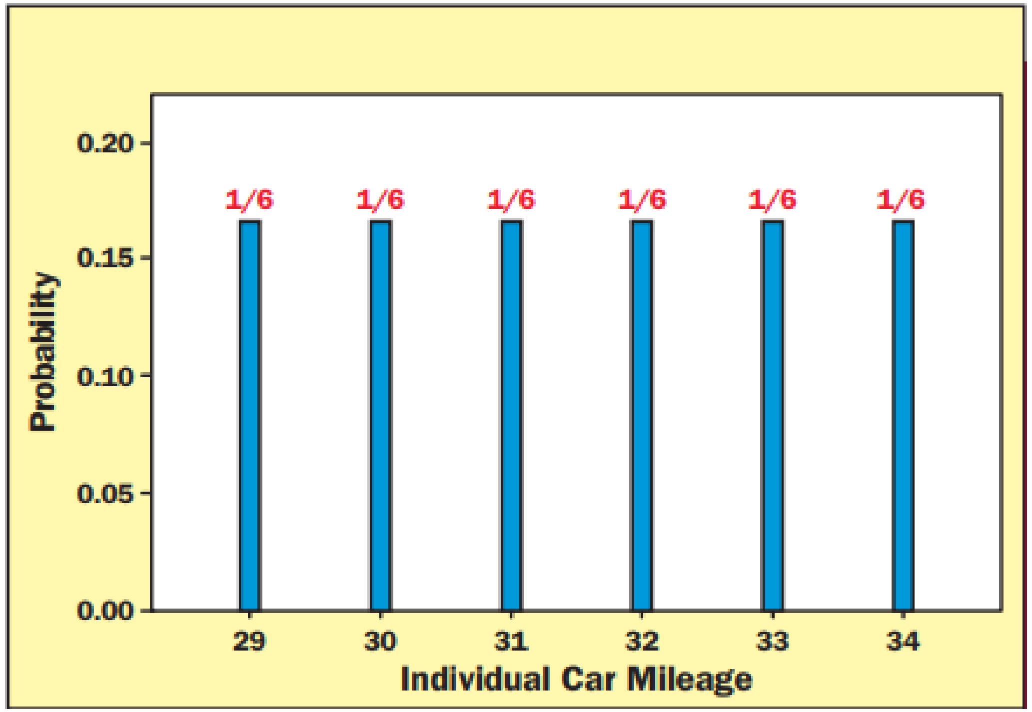
TABLE 7.2

A Probability Distribution Describing the Population of Six Individual Car Mileages

Individual Car Mileage	29	30	31	32	33	34
Probability	1/6	1/6	1/6	1/6	1/6	1/6

FIGURE 7.1 A Comparison of Individual Car Mileages and Sample Means

(a) A graph of the probability distribution describing the population of six individual car mileages



- Different samples of two cars and corresponding mileages would have given different sample means.
- There are, in total, 15 samples of two mileages that could have been obtained by randomly selecting two cars from the population of six cars and subjecting the cars to the mileage test.
- These samples correspond to the 15 combinations of two mileages that can be selected from the six mileages: 29, 30, 31, 32, 33, and 34. The samples are given, along with their means, in Table 7.3(a).

TABLE 7.3 The Population of Sample Means

(a) The population of the 15 samples of $n = 2$ car mileages and corresponding sample means

Sample	Car Mileages	Sample Mean
1	29, 30	29.5
2	29, 31	30
3	29, 32	30.5
4	29, 33	31
5	29, 34	31.5
6	30, 31	30.5
7	30, 32	31
8	30, 33	31.5
9	30, 34	32
10	31, 32	31.5
11	31, 33	32
12	31, 34	32.5
13	32, 33	32.5
14	32, 34	33
15	33, 34	33.5

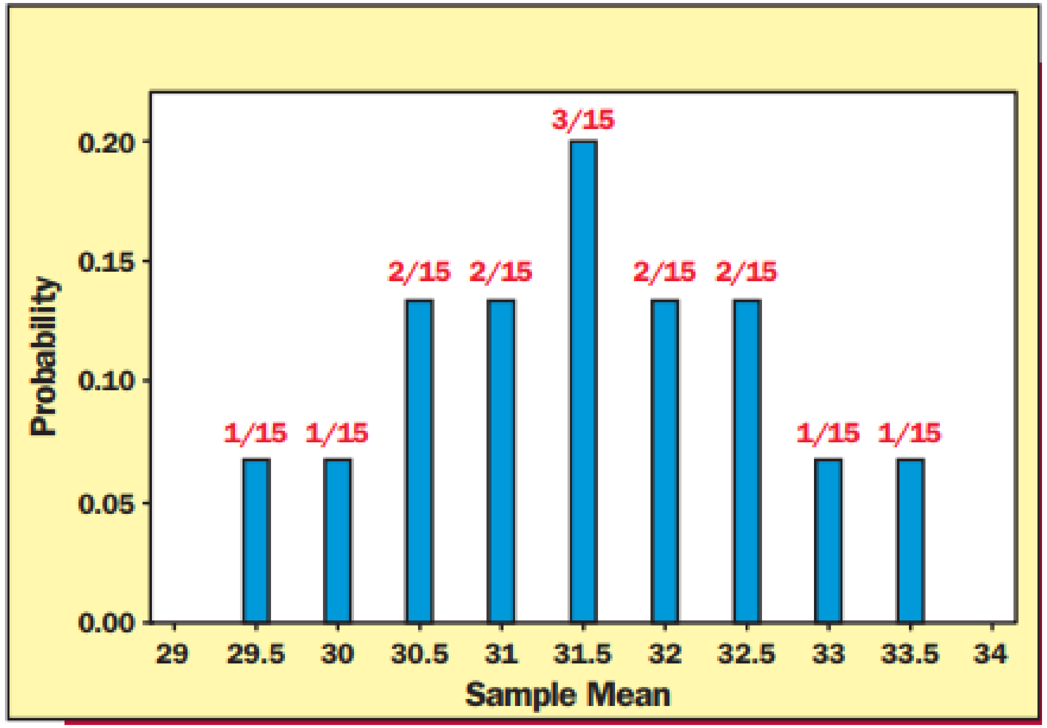
- The probability distribution of the population of sample means is as given in Table 7.3(b). This distribution is *the sampling distribution of the sample mean*.

(b) A probability distribution describing the population of 15 sample means: the sampling distribution of the sample mean

Sample Mean	Frequency	Probability
29.5	1	1/15
30	1	1/15
30.5	2	2/15
31	2	2/15
31.5	3	3/15
32	2	2/15
32.5	2	2/15
33	1	1/15
33.5	1	1/15

- A graph of this distribution is shown in Figure 7.1(b) and illustrates the accuracies of the different possible sample means as point estimates of the population mean. For example, whereas 3 out of 15 sample means exactly equal the population mean of 31.5 mpg, other sample means differ from the population mean by amounts varying from 0.5 mpg to 2 mpg.

(b) A graph of the probability distribution describing the population of 15 sample means



One of the purposes of the sampling distribution of the sample mean is to tell us how accurate the sample mean is likely to be as a point estimate of the population mean

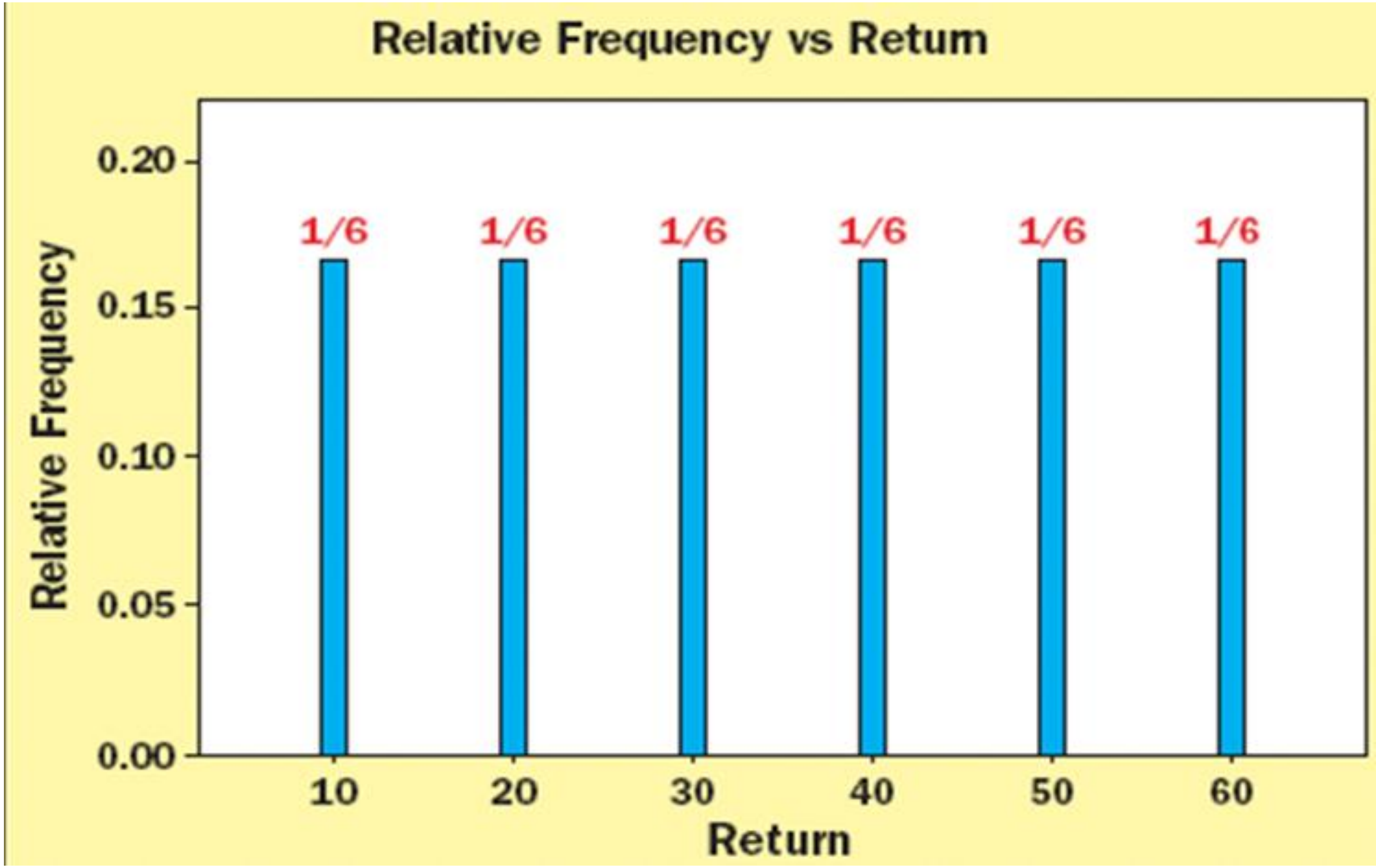
Example: The Stock Return Case

- We have a population of the percent returns from six stocks
  - In order, the values of percentage return are:

    10%, 20%, 30%, 40%, 50%, and 60%

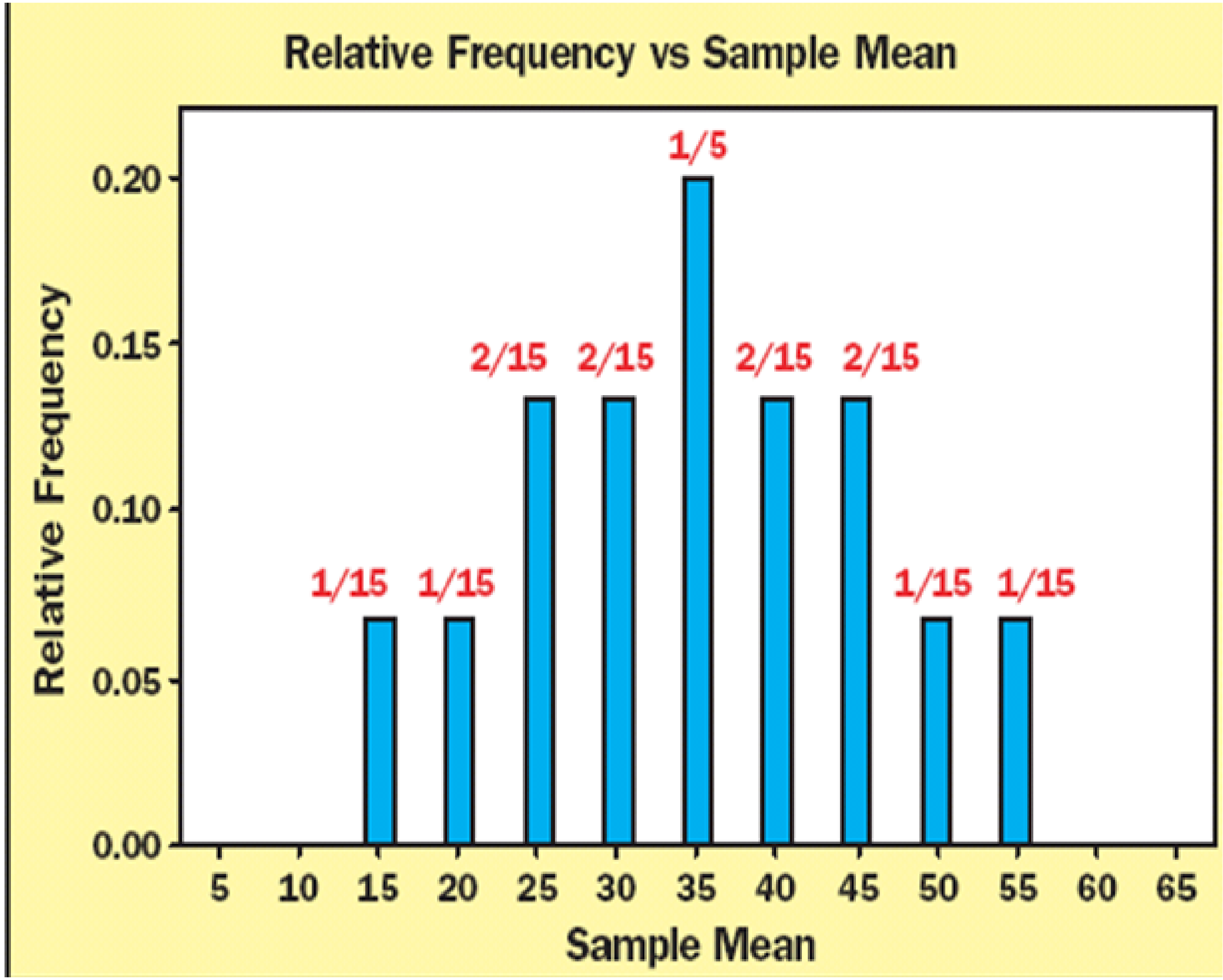
    - Label each stock A, B, C, D, E, F in order of increasing percentage return.
    - The mean rate of return is 35% with a standard deviation of 17.078%.
  - Any one stock of these stocks is as likely to be picked as any other of the six
    - Discrete uniform distribution with $N = 6$.
    - Each stock has a probability of being picked of 1/6.

Stock	% Return	Frequency	Relative Frequency
Stock A	10	1	**1/6**
Stock B	20	1	**1/6**
Stock C	30	1	**1/6**
Stock D	40	1	**1/6**
Stock E	50	1	**1/6**
Stock F	60	1	**1/6**
Total		**6**	**1**


Relative Frequency vs Return
Relative Frequency
0.20
0.15
0.10
0.05
0.00
1/6
1/6
1/6
1/6
1/6
1/6
10
20
30
40
50
60
Return

- Now, select all possible samples of size $n = 2$ from this population of stocks of size $N = 6$
  - That is, select all possible pairs of stocks.
- How to select?
  - Sample randomly.
  - Sample without replacement.
  - Sample without regard to order.
- Finally, calculate the sample means and identify their relative frequencies. Then draw the histogram of the relative frequency (next page).

Sample mean	Frequency	Relative frequency
15	1	1/15
20	1	1/15
25	2	2/15
30	2	3/15
35	3	3/15
40	2	2/15
45	2	2/15
50	1	1/15
55	1	1/15



Observations from this example

- The population of $N = 6$ stock returns has a uniform distribution (slide 31).

- But the histogram of $n = 15$ sample mean returns:

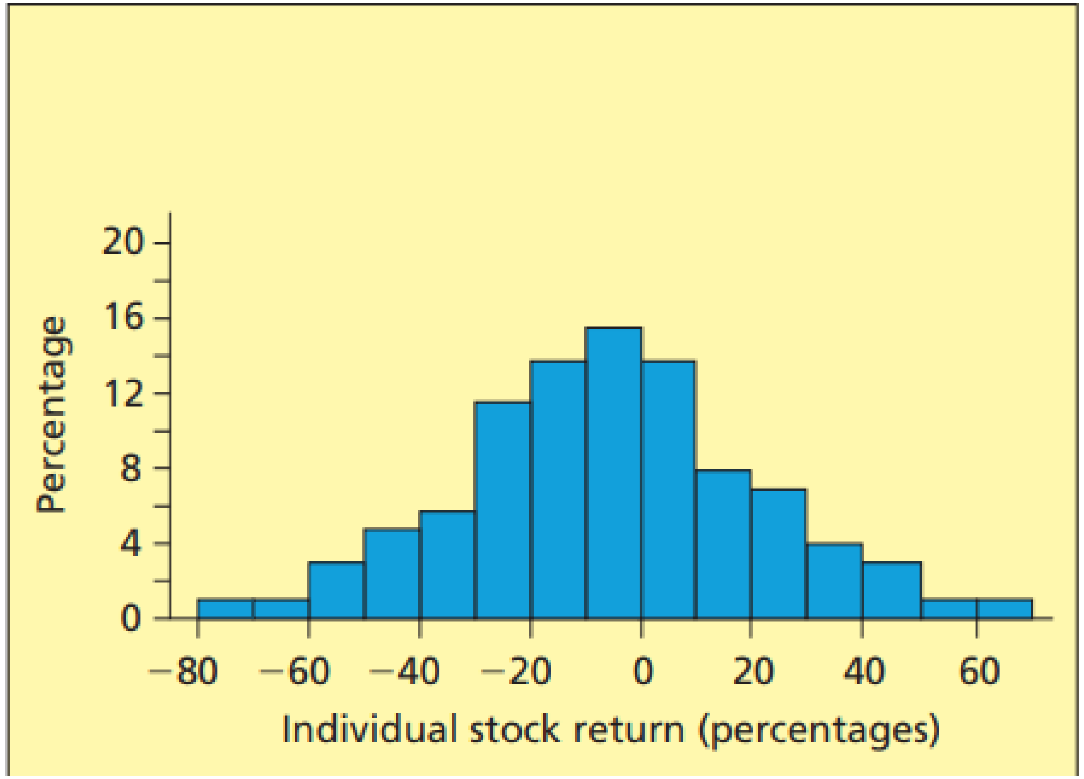
1. Seems to be centered over the same mean return of 35%, and

2. Appears to be bell-shaped and less spread out than the histogram of individual returns.

Example: Sampling all the stocks

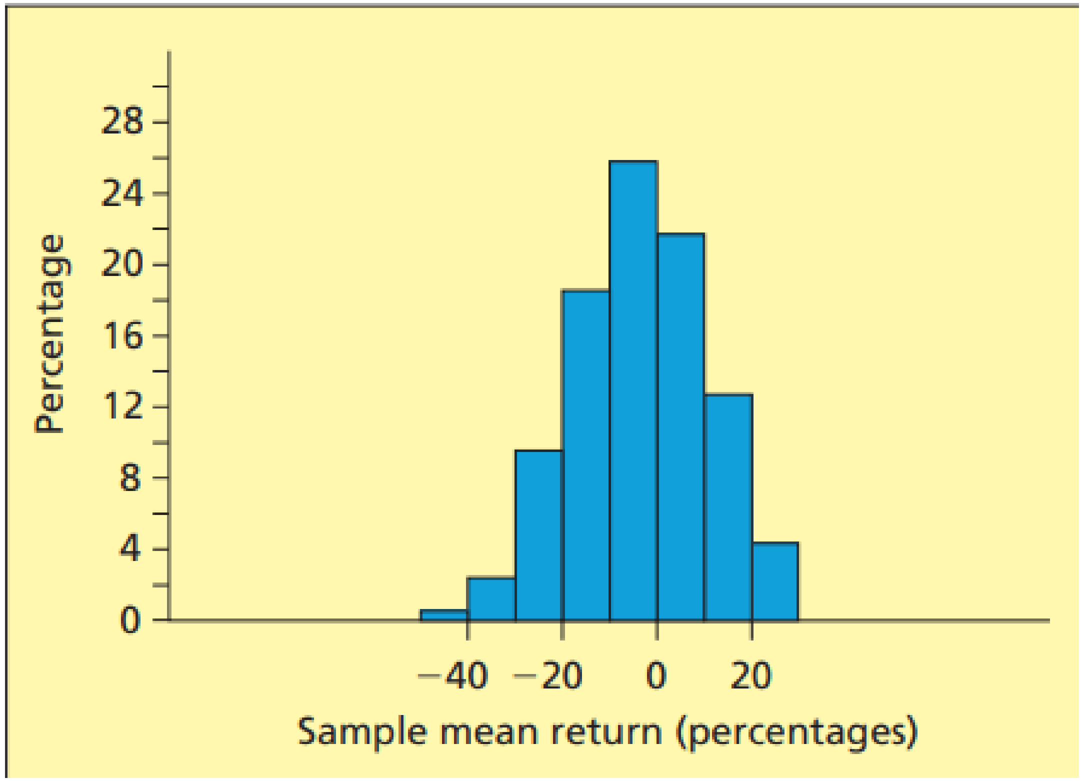
- Consider the population of returns of all 1,815 stocks listed on NYSE for 1987
  - The mean rate of return μ was -3.5% with a standard deviation σ of 0.26.

(a) The percent frequency histogram describing the population of individual stock returns

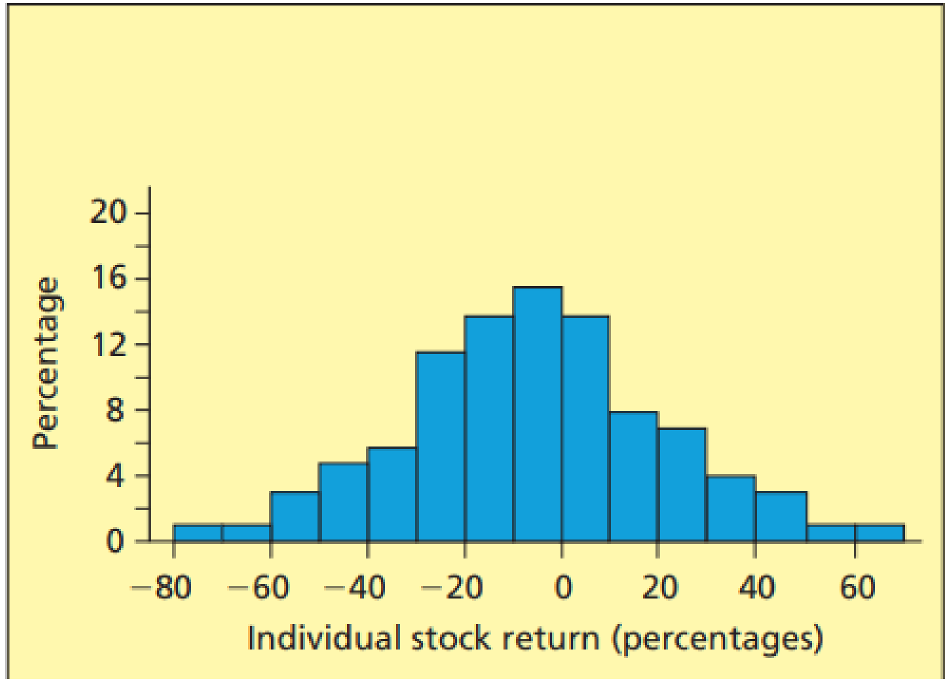


Draw all possible random samples of size $n = 5$ and calculate the sample mean return $\bar{x}$ of each Sample with a computer.

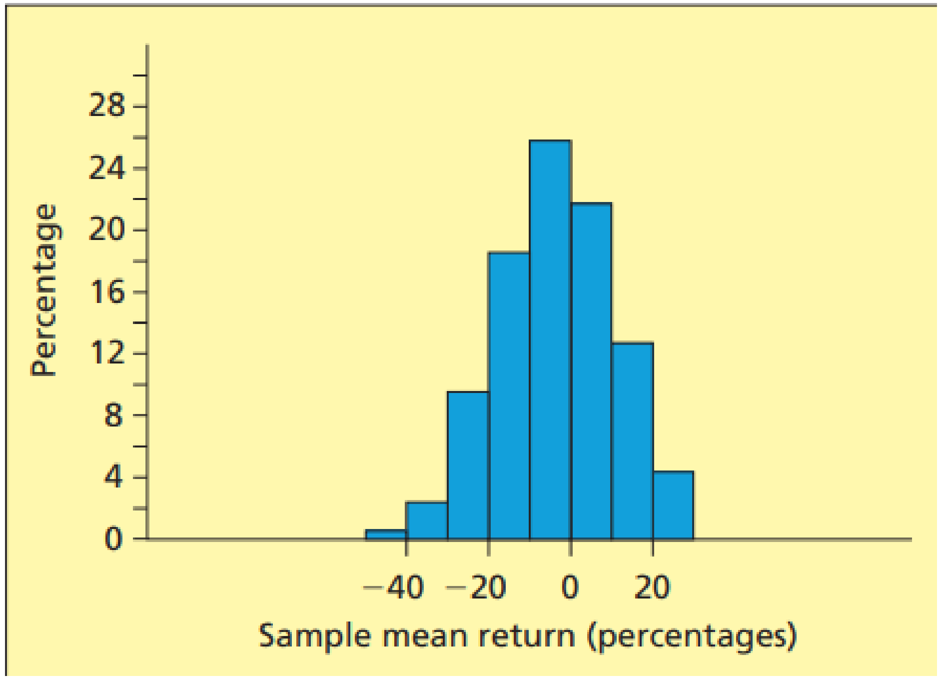
(b) The percent frequency histogram describing the population of all possible sample mean returns when $n = 5$



(a) The percent frequency histogram describing the population of individual stock returns



(b) The percent frequency histogram describing the population of all possible sample mean returns when $n = 5$



- **Observations**
  - Both histograms appear to be bell-shaped and centered over the same mean of -3.5%.
  - The histogram of the sample mean returns looks less spread out than that of the individual returns.

- **Statistics of this example**
  - Mean of all sample means: $\mu_{\bar{x}} = \mu = -3.5.$
  - Standard deviation of all possible means:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.26}{\sqrt{5}} = 0.1163$$

End of Lecture 13

7.2.3 General Conclusions

(1) If the population of individual items is normal distribution, then the population of all sample means is also normal distribution.

(2) Even if the population of individual items is not normal, there are circumstances that the population of all sample means is normal (see *Central Limit Theorem* later)

(3) The mean of all possible sample means equals the population mean

- That is, $\mu_{\bar{x}} = \mu$

(4) The standard deviation $\sigma_{\bar{x}}$ of all sample means is less than the standard deviation of the population.

- That is, $\sigma_{\bar{x}} < \sigma$

The empirical rule holds for the sampling distribution of the sample mean.

- 68.26% of all possible sample means are within (plus or minus) one standard deviation $\sigma_{\bar{x}}$ of $\mu_{\bar{x}}$

- 95.44% of all possible observed values of *x* are within (plus or minus) two $\sigma_{\bar{x}}$ of $\mu_{\bar{x}}$
  - In the example, 95.44% of all possible sample mean returns are in the interval [-3.5 $\pm$ (2×11.63)] = [-3.5 $\pm$ 23.26]
  - That is, 95.44% of all possible sample means are between -26.76% and 19.76%

- 99.73% of all possible observed values of *x* are within (plus or minus) three $\sigma_{\bar{x}}$ of $\mu_{\bar{x}}$.

7.2.4 Properties of the Sampling Distribution of the Sample Mean

(1) The mean $\mu_{\bar{x}}$ of the sampling distribution of $\bar{x}$ is $\mu_{\bar{x}} = \mu$

That is, the mean of all possible sample means is the same as the population mean. This holds for any population size, for any population distribution, and for any sample size.

(2) If the population being sampled is normal, then so is the sampling distribution of the sample mean $\bar{x}$.

(3) If the population being sampled is normal, the variance $\sigma_{\bar{x}}^2$ of the sampling distribution of $\bar{x}$ is

- directly proportional to the variance of the population, and
- inversely proportional to the sample size.

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$$

- Hold exactly if the sampled population (N) is infinite.
- Hold approximately under some conditions if the sampled population is finite (e.g., if the sampled population is much larger than (say, at least 20 times) the size of the sample).

(4) $\bar{x}$ is the point estimate of μ, and the larger the sample size n, the more accurate the estimate, because when n increases, $\sigma_{\bar{x}}$ decreases, $\bar{x}$ is more clustered to the population.

- In order to reduce $\sigma_{\bar{x}}$, take bigger samples!

(5) The standard deviation $\sigma_{\bar{x}}$ of the sampling distribution of $\bar{x}$ follows, in theory, from $\sigma_{\bar{x}}^2$. The formula for $\sigma_{\bar{x}}$ is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

- There is an exact formula for $\sigma_{\bar{x}}$ when the sampled population is finite. However, this is beyond the scope of this course. It is important to use this exact formula if the sampled population is finite and less than 20 times the size of the sample.

For example, if the automaker produces 100,000 new cars this year, and if we randomly select a sample of n = 50 of these cars, then the population size of 100,000 is more than 20 times the sample size of 50 (which is 1,000).

We will assume that all of the remaining populations to be discussed in this book are at least 20 times the size of the sample. Therefore, it will be appropriate to use the formula $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

Example 7.3: The Car Mileage Case: Estimating mean mileage

Part 1: Basic concepts

- Consider the infinite population of the mileages of all of the new cars.
- Assume that this population is normally distributed with mean μ and standard deviation $\sigma = 0.8$ (see Figure 7.3(a)).
- The population of all possible sample means is normally distributed with mean $\mu_{\bar{x}} = \mu$ and standard deviation $\sigma_{\bar{x}} = \sigma/\sqrt{n} = 0.8/\sqrt{n}$.
- In order to show that a larger sample is more likely to give a more accurate point estimate $\bar{x}$ of μ, compare taking a sample of size $n = 5$ with taking a sample of size $n = 50$.

- If $n = 5$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{5}} = 0.358$$

By the Empirical Rule, 95.44 percent of all possible sample means are within $[\mu - 2\sigma_{\bar{x}}, \mu + 2\sigma_{\bar{x}}] = [\mu - 0.716, \mu +$

- If $n = 50$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{50}} = 0.113$$

95.44 percent of all possible sample means are within $[\mu - 2\sigma_{\bar{x}}, \mu + 2\sigma_{\bar{x}}] = [\mu - 0.226, \mu + 0.226]$ mpg.

This implies that the larger sample of size $n = 50$ is more likely to give a sample mean $\bar{x}$ that is near μ (see Figures 7.3(b) and (c)).

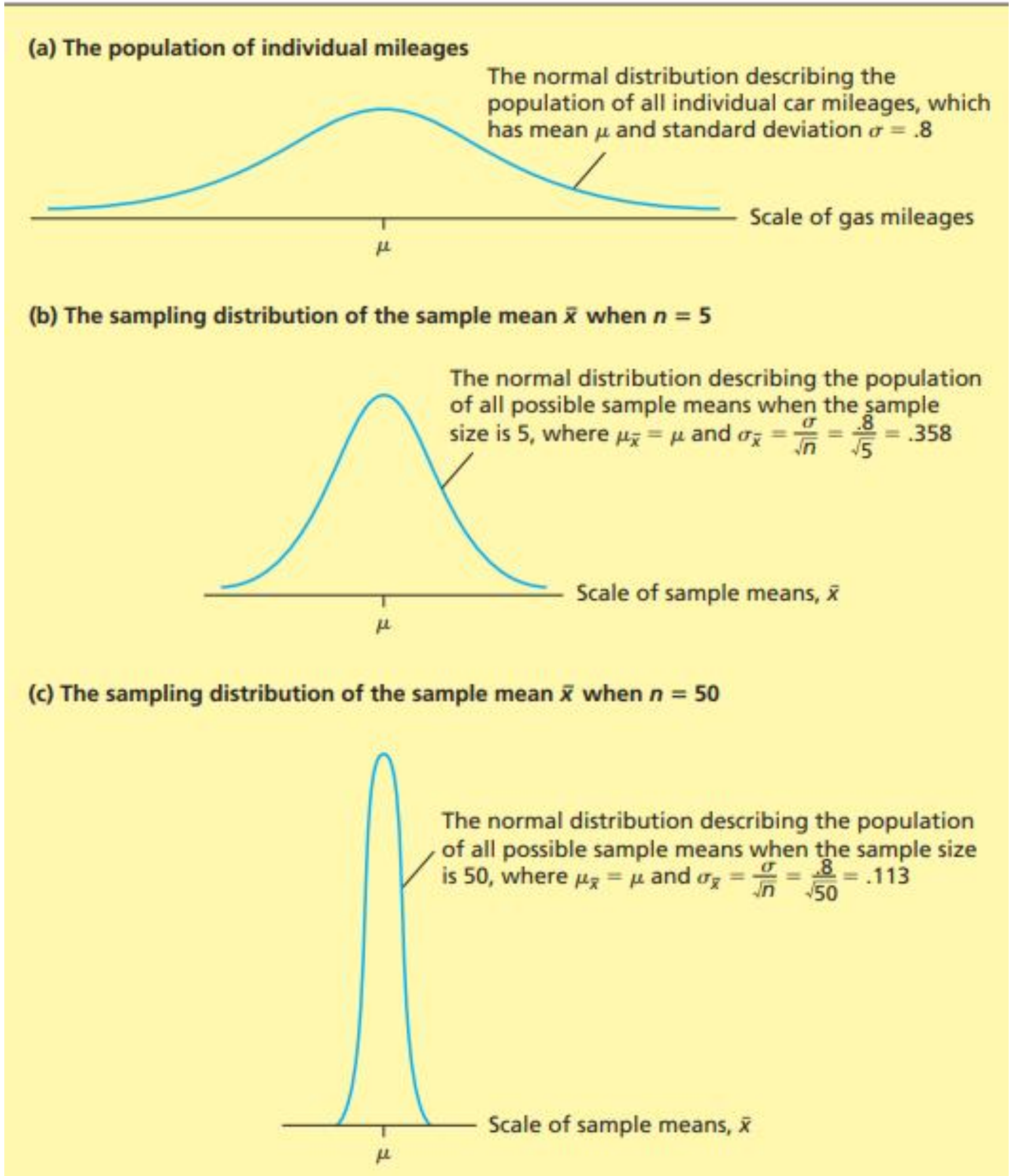


Figure 7.3. A Comparison of (a) the population of all individual car mileages, (b) the Sampling distribution of the Sample mean when *n* = 5, and (c) the Sampling Distribution of the Sample Mean when *n* = 50.

Part 2: : Statistical inference

- Recall from Chapter 3 that the automaker has randomly selected a sample of $n = 50$ mileages, which has mean $\bar{x} = 31.56$.

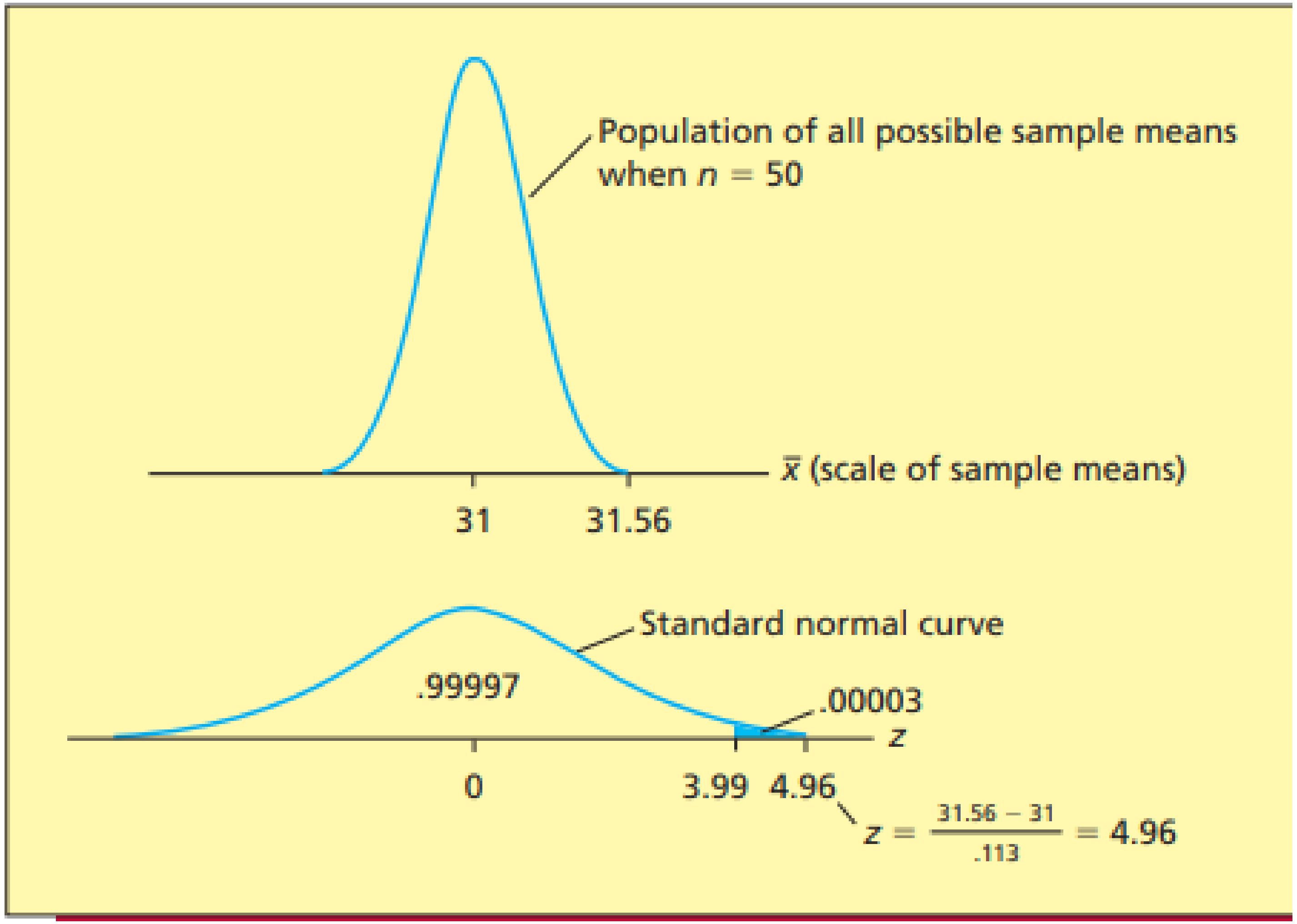
- We now ask the following question: If the population mean mileage μ exactly equals 31 mpg, what is the probability of observing a sample mean mileage that is greater than or equal to 31.56 mpg?

- To find this probability, recall from Chapter 2 that a histogram of the 50 mileages indicates that the population of all individual mileages is normally distributed.
- Assuming that the population standard deviation σ is known to equal 0.8 mpg, it follows that the sampling distribution of the sample mean $\bar{x}$ is a normal distribution, with mean $\mu_{\bar{x}} = \mu$ and standard deviation $\sigma_{\bar{x}} = \sigma/\sqrt{n} = 0.8/\sqrt{50} = 0.113.$
- Therefore,

$$P(\bar{x} \geq 31.56 \text{ if } \mu = 31) = P\left(z \geq \frac{31.56 - \mu_{\bar{x}}}{\sigma_{\bar{x}}}\right)$$

$$= P\left(z \geq \frac{31.56 - 31}{0.113}\right) = P(z \geq 4.96) < 0.003 \text{ (see the figure of next page)}$$

FIGURE 7.4 The Probability That $\bar{x} \geq 31.56$ When $\mu = 31$ in the Car Mileage Case



To find $P(z \geq 4.96)$, you can use z value table in Table A.3 (page 791).

- Therefore, if we are to believe that μ equals 31, then we must believe that we have observed a sample mean that can be described as a smaller than 3 in 100,000 chance.
- Because it is extremely difficult to believe that such a small chance would occur.
- We have extremely strong evidence that μ does not equal 31 and that μ is, in fact, larger than 31.
- This evidence would probably convince the federal government that the model's mean mileage μ exceeds 31 mpg and thus that the model deserves the tax credit.

7.2.5 Central Limit Theorem

➢ If the population being sampled is normal, then the sample mean $\bar{x}$ is also normal.

➢ If the population is non-normal, what is the shape of the sampling distribution of the sample means?

➢ In fact, the sampling distribution is approximately normal if the sample size is large enough, even if the population is non-normal.

- The population of all possible sample means is ***approximately*** normal with

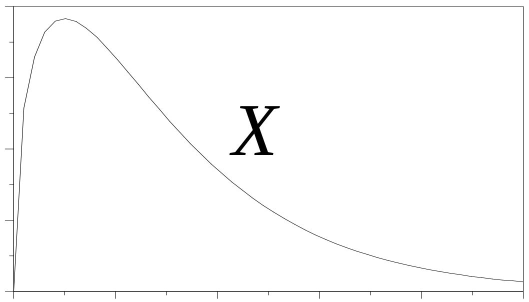
$$\boxed{\mu_{\bar{x}} = \mu}$$

- No matter what is the probability distribution that describes the population, if the sample size n is large enough, then the population of **all** possible sample means is ***approximately*** normal with standard deviation

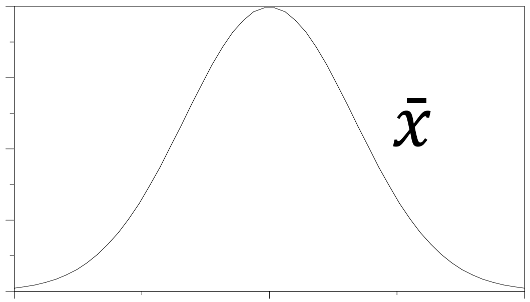
$$\boxed{\sigma_{\bar{x}} = \sigma/\sqrt{n}}$$

- Further, the larger the sample size n, the closer the sampling distribution of the sample means is to being normal
    - In other words, the larger n, the better the approximation.

Random Sample $(x_1, x_2, \ldots, x_n)$



as $n \to$ large



Population Distribution

(μ, σ)

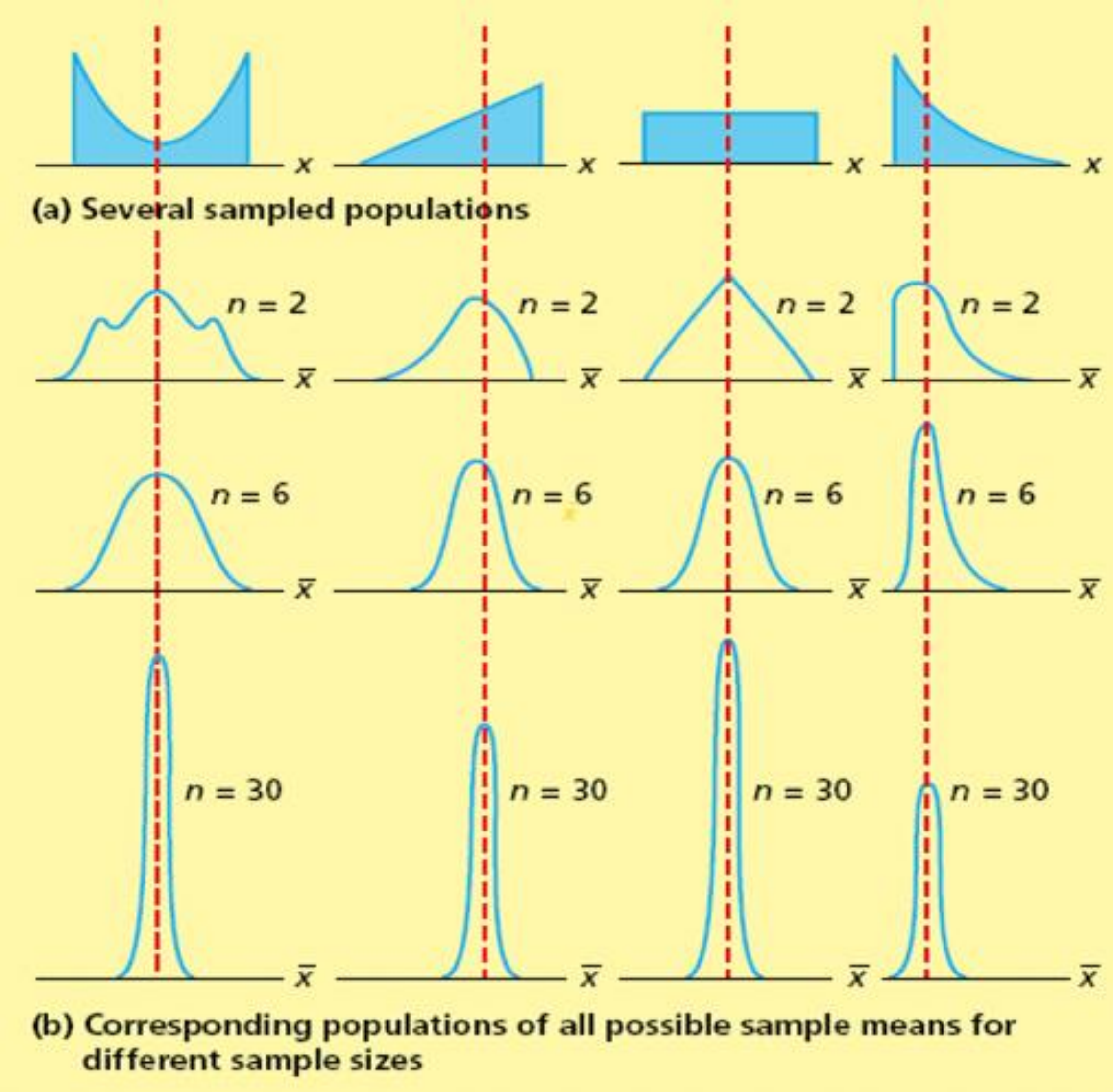
(right-skewed)

Sampling Distribution of Sample Means

$(\mu_{\bar{x}} = \mu, \sigma_{\bar{x}} = \sigma/\sqrt{n})$

(nearly normal)

Example: effect of the Sample Size



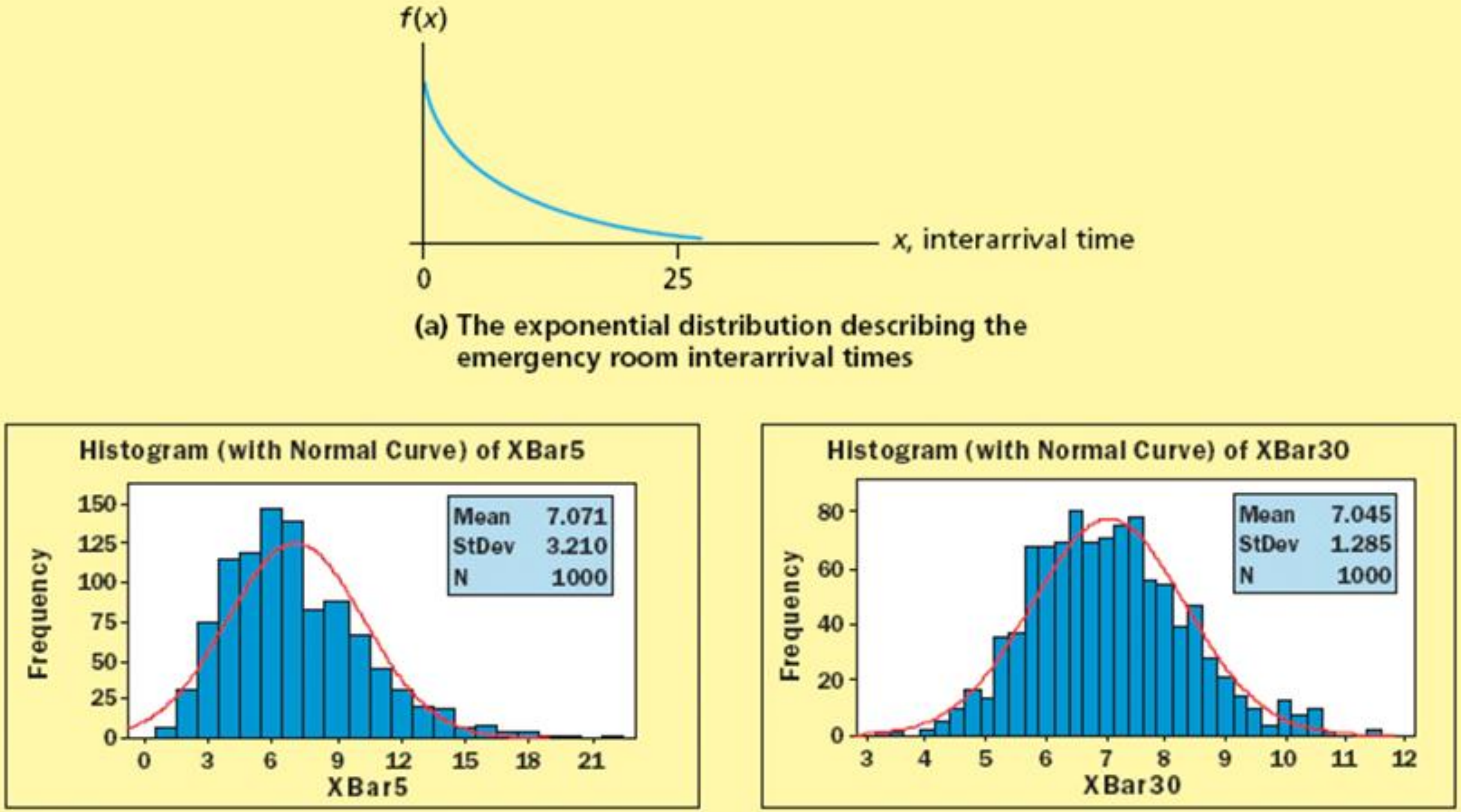
The larger the sample size, the more nearly normally distributed is the population of all possible sample means.

Also, as the sample size increases, the spread of the sampling distribution decreases.

How Large?

- How large is “large enough?”
- If the sample size n is at least 30, then for most sampled populations, the sampling distribution of sample means is approximately normal.
- If the population is normal, the sampling distribution of sample means $\bar{x}$ is normal regardless of the sample size.

- Refer to Figure 7.6 on next slide
  - Shown in Fig 7.6(a) is an exponential (right skewed) distribution
  - In Figure 7.6(b), 1,000 samples of size $n = 5$
    - Slightly skewed to right
  - In Figure 7.6(c), 1,000 samples with $n = 30$
    - Approximately bell-shaped and normal



(a) The exponential distribution describing the emergency room interarrival times

(b) A histogram of 1,000 sample means based on samples of size 5

(c) A histogram of 1,000 sample means based on samples of size 30

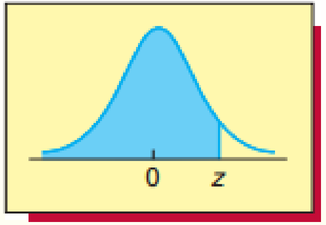
Example A:

The foreman of a bottling plant has observed that the amount of soda in each "32-ounce" bottle (bo·tl) is actually a normally distributed random variable, with a mean of 32.2 ounces and a standard deviation of 0.3 ounce.

If a customer buys one bottle, what is the probability that the bottle will be greater than 32 ounces?

Solution:

TABLE A.3 Cumulative Areas under the Standard Normal Curve (*concluded*)



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852

Example B:

The foreman of a bottling plant has observed that the amount of soda in each "32-ounce" bottle is actually a normally distributed random variable, with a mean of 32.2 ounces and a standard deviation of 0.3 ounce.

If a customer buys a carton of four bottles, what is the probability that the *mean amount of the four bottles* will be greater than 32 ounces?

Solution:

X is normally distributed with μ =32.2 and σ = 0.3.

X is normally distributed, therefore so will $\bar{x}$.

$$\mu_{\bar{x}} = \mu = 32.2 \text{ oz.}$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.3}{\sqrt{4}} = 0.15$$

We want to find $P(\bar{x} > 32)$,

$$P(\bar{x} > 32) = P\left(\frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} > \frac{32 - 32.2}{0.15}\right) = P(Z > -1.33\ ) = 0.9082$$

There is about a 91% chance the mean of the four bottles will exceed 32 oz.

Example

A light bulb manufacturer claims that the lifespan of its light bulbs has a mean of 54 months and a standard deviation of 6 months. Your consumer advocacy group tests 50 of them. Assuming the manufacturer's claims are true, what is the probability that it finds a mean lifetime of 52 months or less?

Solution:

$\bar{x}$ is approximately normally distributed.

$$\mu_{\bar{x}} = \mu, \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

We are seeking $P(\bar{x} \leq 52)$.

Convert to Z-scores (Standard normal distribution)

$$z = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{52 - 54}{0.85} = -2.35$$

Thus, the probability of 52 months or less is 0.0094, or 0.94%.

Example 7.4 The e-billing Case: Reducing Mean Bill Payment Time

Recall that a management consulting firm has installed a new computer-based electronic billing system in a Hamilton, Ohio, trucking company. The management consulting firm believes that the new system will reduce the mean bill payment time by more than 50 percent. The mean payment time using the old billing system was approximately equal to, but no less than, 39 days. Therefore, if μ denotes the new mean payment time, the consulting firm believes that μ will be less than 19.5 days.

Moreover, the firm has installed electronic billing systems for clients in other industries. Analysis of results from these installations shows that, although the population mean payment time μ varies from company to company, the population standard deviation σ of payment times is the same for different applications and equals 4.2 days.

To assess whether μ is less than 19.5 days, the consulting firm has randomly selected a sample of n = 65 invoices processed using the new billing system and has determined the payment times for these invoices. The mean of the 65 payment times is $\bar{x} =$ 18.1077 days, which is less than 19.5 days.

Therefore, we ask the following question: If the population mean payment time is 19.5 days, what is the probability of observing a sample mean payment time that is less than or equal to 18.1077 days?

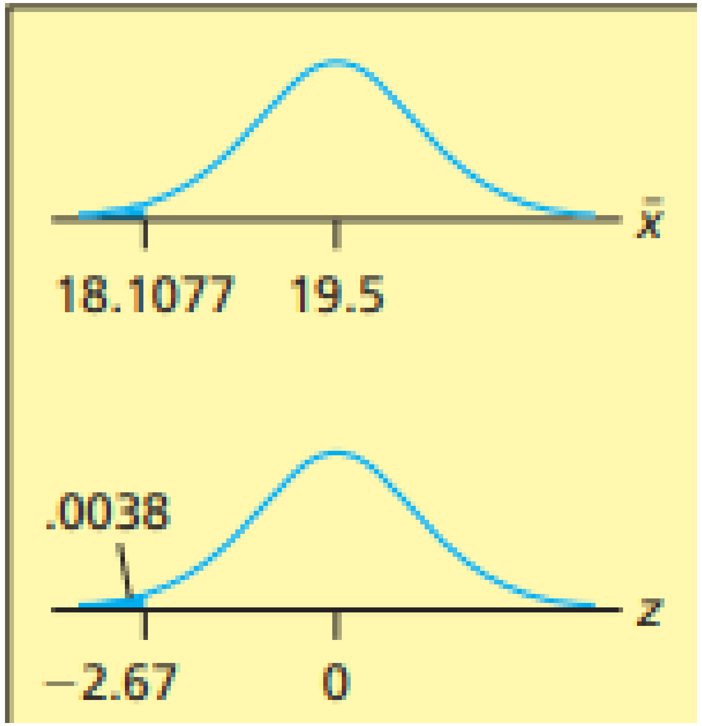
Solution:

The Central Limit Theorem tells us that, because the sample size n = 65 is large, the sampling distribution of is approximately a normal distribution with mean $\mu_{\bar{x}} = \mu$ and standard deviation $\sigma_{\bar{x}} = \sigma/\sqrt{n}$.

Assuming that σ also equals 4.2 days, it follows that $\sigma_{\bar{x}}$ equals $4.2/\sqrt{65} = 0.5209$ and that

$$P(\bar{x} \leq 18.1077 \text{ if } \mu = 19.5) = P\left(z \leq \frac{18.1077 - 19.5}{.5209}\right) = P(z \leq -2.67)$$

which is the area under the standard normal curve to the left of -2.67. The normal table tells us that this area equals .0038. This probability says that, if μ equals 19.5, then only 0.0038 of all possible sample means are at least as small as the sample mean $\bar{x} = 18.1077$ that we have actually observed. Therefore, if we are to believe that μ equals 19.5, we must believe that we have observed a sample mean that can be described as a 38 in 10,000 chance. It is very difficult to believe that such a small chance would occur, so we have very strong evidence that μ does not equal 19.5 and is, in fact, less than 19.5. We conclude that the new billing system has reduced the mean bill payment time by more than 50 percent.



End of Lecture 14

Review of key ideas

The mean of the sample means is $\mu_{\bar{x}} = \mu$. No matter what the distribution of the sampled population is, no matter how big N and n are.

If the distribution of the sampled population is normal, the distribution of **All** sample means is normal. No matter how big are N and n.

If the distribution of the sampled population is normal and N is big (e.g., $N \geq 20n$), $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$.

If n is big (e.g., $n \geq 30$), the distribution of **All** sample means is approximately normal with $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$. No matter what the distribution of the sampled population is, no matter how big N is.

If *N* is not large enough?

Need to consider

- sampling without replacement, and
- sampling with replacement, separately.

7.2.6 Technical Note:

If we randomly select a sample of size n without replacement from a finite population of size N, then it can be shown that $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\frac{N-n}{N-1}$, where the quantity $\sqrt{\frac{N-n}{N-1}}$ is called the finite population multiplier (FPM). If the size of the sampled population is at least 20 times the size of the sample (that is, if $N \geq 20n$), then the finite population multiplier is approximately equal to one, and $\sigma_{\bar{x}}^2$ approximately equals $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$. However, if the population size N is smaller than 20 times the size of the sample, then the finite population multiplier is substantially less than one, and we must include this multiplier in the calculation of $\sigma_{\bar{x}}^2$.

We will see how the formula $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\frac{N-n}{N-1}$ can be used to make statistical inferences in Section 8.5.

- Finite population multiplier (FPM) must be used if:

  (1) sampling without replacement, and

  (2) the sampled population (N) is small comparing to the sample size (n) (e.g., $\frac{N}{n} < 20$).

- Finite population multiplier (FPM) can be ignored if:

  the sampled population (N) is large comparing to the sample size (n) (e.g., $\frac{N}{n} > 20$).

  In this situation, sampling with replacement and sampling without replacement make little difference. This is, sampling without replacement can be regarded as a good approximation of sampling with replacement and $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$ is a sufficiently good approximation.

Verification Example:

There are 3 apples in a basket ($N = 3$). Their values of weight are 100 g, 150 g and 200 g, respectively. Take 2 apples every time ($n = 2$). We obtain the following samples:

(Case 1: with replacement)

Sample number	Partners	Mean
1	100, 100	100
2	100, 150	125
3	100, 200	150
4	150, 100	125
5	150, 150	150
6	150, 200	175
7	200, 100	150
8	200, 150	175
9	200, 200	200

(Case 2: without replacement)

Sample number	Partners	Mean
1	100, 150	125
2	100, 200	150
3	150, 100	125
4	150, 200	175
5	200, 100	150
6	200, 150	175

We calculate:

(1) the mean and the variance of the population

$$\mu = \frac{100 + 150 + 200}{3} = 150$$

$$\sigma^2 = \frac{(100 - 150)^2 + (150 - 150)^2 + (200 - 150)^2}{3} = 1666.67$$

(2) the mean and the variance of the populations of sample means

(Case 1: with replacement)

$$\mu_{\bar{x}} = \frac{100+125+150+125+150+175+150+175+200}{9} = 150$$

$$\sigma_{\bar{x}}^2 = \frac{(100-150)^2+(125-150)^2+(150-150)^2+(125-150)^2+(150-150)^2+(175-150)^2+(150-150)^2+(175-150)^2+(200-150)^2}{9}$$

$$= 833.3$$

This confirms:

$$\mu_{\bar{x}} = \mu, \qquad \sigma_{\bar{x}}^2 = \frac{\sigma^2}{2} = \frac{\sigma^2}{n}$$

(Case 2: without replacement)

$$\mu_{\bar{x}} = \frac{125 + 150 + 125 + 175 + 150 + 175}{6} = 150$$

$$\sigma_{\bar{x}}^2 = \frac{(125-150)^2 + (150-150)^2 + (125-150)^2 + (175-150)^2 + (150-150)^2 + (175-150)^2}{6} = 416.67$$

This confirms:

$$\mu_{\bar{x}} = \mu, \quad \sigma_{\bar{x}}^2 = \frac{\sigma^2}{2}\frac{3-2}{3-1} = \frac{\sigma^2}{n}\frac{N-n}{N-1}$$

In this example, N is only 1.5 times the sample size n and the finite population multiplier (FPM) is very significant.

7.2.7 Unbiasedness and minimum-variance estimates

Recall that a sample statistic is any descriptive measure of the sample measurements. For instance, the sample mean $\bar{x}$ is a statistic, and so are the sample median, the sample variance s^2 , and the sample standard deviation s. Not only do different samples give different values of $\bar{x}$, different samples also give different values of the median, s^2, s, or any other statistic. It follows that, before we draw the sample, any sample statistic is a random variable, and

> The sampling distribution of a sample statistic is the probability distribution of the population of all possible values of the sample statistic.

In general, we wish to estimate a population parameter by using a sample statistic that is what we call an unbiased point estimate of the parameter.

> A sample statistic is an unbiased point estimate of a population parameter if the mean of the population of all possible values of the sample statistic equals the population parameter.

Unbiased Estimates

- A sample statistic is an ***unbiased*** point estimate of a population parameter if the mean of all possible values of the sample statistic equals the population parameter

- $\bar{x}$ is an unbiased estimate of μ because $\mu_{\bar{x}} = \mu$
  - In general, the sample mean is always an unbiased estimate of μ (In words, the average of all the different possible sample means (that we could obtain from all the different possible samples) equals μ)
  - The sample median is often an unbiased estimate of μ
    - But not always

- If the sampled population is infinite (i. e, $N = \infty$), the sample variance s^2 is an unbiased estimate of σ^2(that is, the average of all the different possible sample variances that we could obtain (from all the different possible samples) is equal to σ^2.)

- That is why s^2 has a divisor of $n - 1$

(if we used n as the divisor when estimating s^2, we would not obtain an unbiased estimate)

- If the sampled population is finite, s^2 may be regarded as an approximately unbiased estimate of σ^2 as long as the population is fairly large (which is usually the case).

However, s is not an unbiased estimate of σ

- Even so, since there is no easy way to calculate an unbiased point estimate of σ, the usual practice is to use s as an estimate of σ.

Example (sampling with replacement)

- Let's consider a small population: {2, 5, 11}

  Population parameters:

  Population size: $N = 3$

  Population mean: $\mu = \frac{2+5+11}{3} = 6$

  Population variance: $\sigma^2 = \frac{(2-6)^2+(5-6)^2+(11-6)^2}{3} = 14$

- All Possible Samples of Size $n = 2$

Let's take all possible samples of size 2 (with replacement)

Sample	Values	Sample mean ($\overline{x}$)	Sample variance (s^2)
1	(2, 2)	2	$\frac{(2-2)^2+(2-2)^2}{n-1}=0$
2	(2, 5)	3.5	4.5
3	(2, 11)	6.5	40.5
4	(5, 2)	3.5	4.5
5	(5, 5)	5	0
6	(5, 11)	8	18
7	(11, 2)	6.5	40.5
8	(11, 5)	8	18
9	(11, 11)	11	0

- Calculate Mean of All Sample Means: Mean of $\bar{x} = \frac{2+3.5+6.5+3.5+5+8+6.5+8+11}{9} = 6$

  This equals our population mean of $\mu = 6$

- Calculate Mean of All Sample Variances

  Mean of s^2 is $\overline{s^2} = \frac{0+4.5+40.5+4.5+0+18+40.5+18+0}{9} = 14$

  This also equals our population variance of $\sigma^2 = 14$

Now, for the same example but sampling without replacement)

- Let's consider the same small population: {2, 5, 11}

  Population parameters:

  Population size: $N = 3$

  Population mean: $\mu = \frac{2+5+11}{3} = 6$

  Population variance: $\sigma^2 = \frac{(2-6)^2+(5-6)^2+(11-6)^2}{3} = 14$

- All Possible Samples of Size n = 2

Let's take all possible samples of size 2 (without replacement)

Sample	Values	Sample mean ($\overline{x}$)	Sample variance (s^2)
1	(2, 5)	3.5	$\frac{(2-3.5)^2+(5-3.5)^2}{n-1} = 4.5$
2	(2, 11)	6.5	40.5
3	(5, 11)	8	18

- Calculate Mean of All Sample Means: Mean of $\bar{x} = \frac{3.5+6.5+8}{3} = 6$

  This equals our population mean of variance of $\mu = 6$

- Calculate Mean of All Sample Variances

  Mean of s^2 is $\overline{s^2} = \frac{4.5+40.5+18}{3} = 21$

  - $\overline{s^2}$ does not equal our population variance of $\sigma^2 = 14.$
  - $\overline{s^2}$ equal to $\frac{N}{N-1}\sigma^2 = \frac{3}{2}(14)$

Hence, when sampling with replacement,

- the formula $\bar{x} = \sum \frac{x_i}{n}$ is an unbiased estimator of the population mean.
  - The mean of all possible sample means $\bar{x}$ equals μ

- the formula $s^2 = \sum \frac{(x_i - \bar{x})^2}{n-1}$ is an unbiased estimator of the population variance.

(uses Bessel's correction (dividing by $n - 1$ instead of n) to make it an unbiased estimator of the population variance.)

  - The mean of all possible sample variance s^2 equals σ^2

These statements hold for any population size *N*.

Hence, when sampling without replacement,

- the formula $\bar{x} = \sum \frac{x_i}{n}$ is still an unbiased estimator of the population mean.
  - The mean of all possible sample means $\bar{x}$ equals μ

- the formula $s^2 = \sum \frac{(x_i - \bar{x})^2}{n-1}$ is NOT an unbiased estimator of the population variance σ^2 but is an unbiased estimator of $\frac{N}{N-1}\sigma^2$.
  - The mean of all possible sample variance s^2 equals $\frac{N}{N-1}\sigma^2$.

- These statements hold for any population size *N*.

- If population N is large enough, sampling with or without replacement results in same estimator of population variance σ^2.

For example, China has 1.4 billion people, which is extremely large. In this case, sampling a 10,000 samples with or without replacement make no difference.

Exercises (if time permits)

7.9 Suppose that we will take a random sample of size n from a population having mean μ and standard deviation σ. For each of the following situations, find the mean, variance, and standard deviation of the sampling distribution of the sample mean $\bar{x}$:

a $\mu = 10, \quad \sigma = 2, \quad n = 25$

b $\mu = 500, \quad \sigma = .5, \quad n = 100$

c $\mu = 3, \quad \sigma = .1, \quad n = 4$

d $\mu = 100, \quad \sigma = 1, \quad n = 1{,}600$

Answer:

a. 10, .16, .4
b. 500, .0025, .05
c. 3, .0025, .05
d. 100, .000625, .025

$$\mu_{\bar{x}} = \mu, \qquad \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$$

End of Lecture 15

Examples (if time permits)

7.11 Suppose that we will randomly select a sample of 64 measurements from a population having a mean equal to 20 and a standard deviation equal to 4.

a Describe the shape of the sampling distribution of the sample mean $\bar{x}$. Do we need to make any assumptions about the shape of the population? Why or why not?

b Find the mean and the standard deviation of the sampling distribution of the sample mean $\bar{x}$.

c Calculate the probability that we will obtain a sample mean greater than 21; that is, calculate $P(\bar{x} > 21)$. Hint: Find the z value corresponding to 21 by using $\mu_{\bar{x}}$ and $\sigma_{\bar{x}}$ because we wish to calculate a probability about $\bar{x}$. Then sketch the sampling distribution and the probability.

d Calculate the probability that we will obtain a sample mean less than 19.385; that is, calculate $P(\bar{x} < 19.385)$.

Answer:

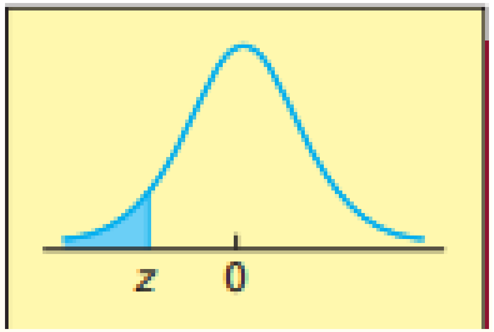
a. Normally distributed
No, sample size is large ($\geq$30)

b. $\mu_{\bar{x}} = 20, \sigma_{\bar{x}} = .5$

c. .0228

d. .1093

T ABLE A.3
Cumulative Areas under the Standard Normal Curve



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
−3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
−3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
−3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
−3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
−3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
−3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
−3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
−3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
−3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379

Examples (if time permits)

7.12 There are six combinations, or samples, of two grand prizes that can be randomly selected from the four grand prizes 20, 40, 60, and 80 thousand dollars. Four of these samples are (20, 40), (20, 60), (20, 80), and (40, 60). Find the other two samples.

7.13 Find the mean of each sample in Exercise 7.12.

Answer:

(40, 80), (60, 80)

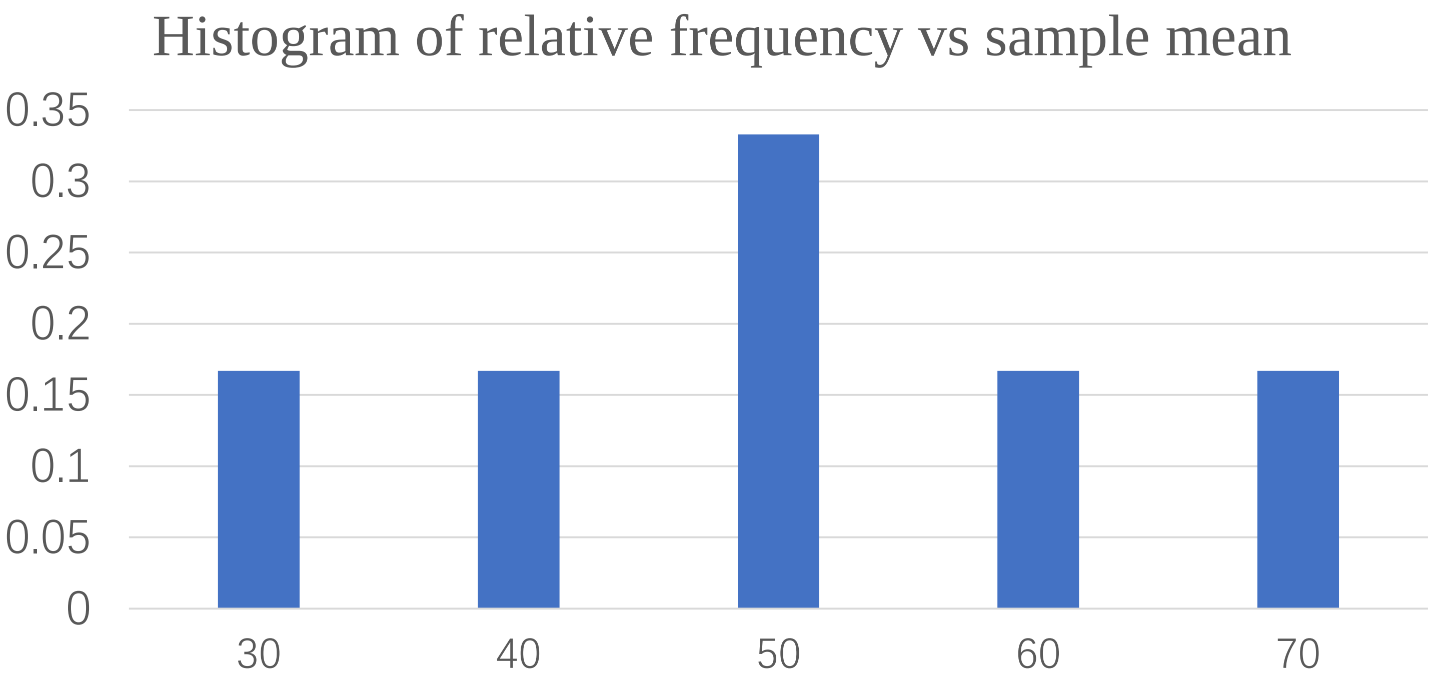
$\bar{x} = 30,40,50,50,60,70$

Examples (if time permits)

7.14 Find the probability distribution of the population of six sample mean grand prizes.

Answer:

Class	Frequency	Relative frequency
30	1	0.167
40	1	0.167
50	2	0.333
60	1	0.167
70	1	0.167



Examples (if time permits)

7.15 If you select two envelopes, what is the probability that you will receive a sample mean grand prize of at least 50 thousand dollars?

Answer:

$$P(\bar{x} \geq 50) = P(\bar{x} = 50) + P(\bar{x} = 60) + P(\bar{x} = 70)$$

$$= 0.333 + 0.167 + 0.167 = 0.667$$

Examples (if time permits)

7.16 Compare the probability distribution of the four individual grand prizes with the probability distribution of the six sample mean grand prizes. Would you select one or two envelopes? Why? Note: There is no one correct answer. It is a matter of opinion.

Answer:

If four envelopes, the probability receiving higher than 50 is 0.5.

So, will select two envelops.

Examples (if time permits)

7.17 THE BANK CUSTOMER WAITING TIME CASE DS WaitTime

Recall that the bank manager wants to show that the new system reduces typical customer waiting times to less than six minutes. One way to do this is to demonstrate that the mean of the population of all customer waiting times is less than 6. Letting this mean be μ, in this exercise we wish to investigate whether the sample of 100 waiting times provides evidence to support the claim that μ is less than 6.

For the sake of argument, we will begin by assuming that μ equals 6, and we will then attempt to use the sample to contradict this assumption in favor of the conclusion that μ is less than 6.

Recall that the mean of the sample of 100 waiting times is $\bar{x} = 5.46$ and assume that σ, the standard deviation of the population of all customer waiting times, is known to be 2.47.

a Consider the population of all possible sample means obtained from random samples of 100 waiting times. What is the shape of this population of sample means? That is, what is the shape of the sampling distribution of $\bar{x}$? Why is this true?

b Find the mean and standard deviation of the population of all possible sample means when we assume that μ equals 6.

c The sample mean that we have actually observed is $\bar{x} = 5.46$. Assuming that μ equals 6, find the probability of observing a sample mean that is less than or equal to $\bar{x} = 5.46$.

d If μ equals 6, what percentage of all possible sample means are less than or equal to 5.46? Because we have actually observed a sample mean of $\bar{x} = 5.46$, is it more reasonable to believe that (1) μ equals 6 and we have observed one of the sample means that is less than or equal to 5.46 when μ equals 6, or (2) that we have observed a sample mean less than or equal to 5.46 because μ is less than 6? Explain. What do you conclude about whether the new system has reduced the typical customer waiting time to less than six minutes?

Answer: $n = 100, \bar{x} = 5.46, \sigma = 2.47$

(a) Normal distributed. because $n = 100 \geq 30$

(b) $\mu_{\bar{x}} = \mu = 6, \sigma_{\bar{x}} = \frac{2.47}{\sqrt{100}} = 0.247$

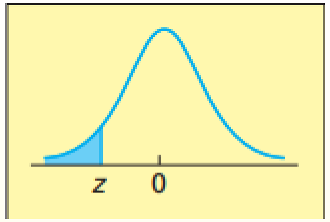
(c)

$$P(\bar{x} \leq 5.46) = P\left(t \leq \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}}\right) =$$

$$P\left(t \leq \frac{5.46 - 6}{0.247}\right) = P(t \leq -2.19) = 0.0143$$

(d) We obtained 1.43% which is small. Such a small probability is unlikely to occur but it occurs. Therefore, the average waiting time is unlikely to be 6 but smaller than 6.

TABLE A.3 Cumulative Areas under the Standard Normal Curve



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

Section 7.3

The Sampling Distribution of the Sample Proportion

Example: There are 200,000 eligible voters (population) in York County, South Carolina. Among all voters, 120,000 voters preferred (said "yes" to) Candidate ***A***. So the proportion of voters preferred Candidate ***A*** is p = 0.6. In this example, p is the population proportion.

However:

- In most situations, obtaining the population proportion is difficulty**.** In words, p is unknown.

Survey:

- Randomly Select 100 York County voters (a sample) and return the proportion $\hat{p}$ of voters preferred (said "yes" to) Candidate ***A*.**
- Select another 100 York County voters (another sample) will return another proportion $\hat{p}$.
- And so forth. Different samples yield different values of sample proportion $\hat{p}$.

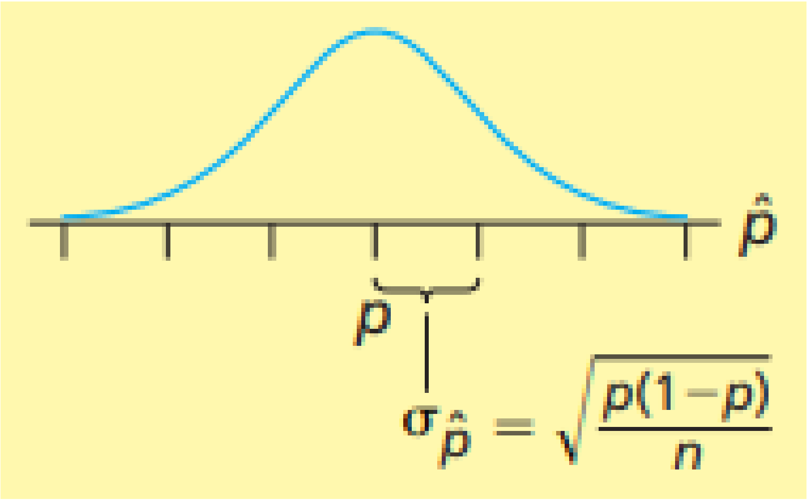
The sampling distribution of the sample proportion describes the distribution of possible sample proportions ($\hat{p}$) that would result from taking many random samples of the same size (n) from a population with a true proportion (p).

- For a population of units, we select samples of size n, and calculate its proportion $\hat{p}$ for the units of the sample to be fall into a particular category.
- $\hat{p}$ is a random variable and has its probability distribution.
- The probability distribution of all possible sample proportions is called the sampling distribution of the sample proportion.

Three key parameters:

- Population proportion, p. Just one number, which is usually unknown
- Size of the samples, n. The sizes of the samples must be same.
- Sample proportions, $\hat{p}$. Different samples have different proportions.

For the population of all possible sample proportions, the sampling distribution of $\hat{p}$ is



(1) approximately normal, if n is large (meet the conditions that $np \geq 5$ and $n(1-p) \geq 5$)

(2) has mean $\mu_{\hat{p}} = p$ (valid for any sample size and $\hat{p}$ is an unbiased estimate of p. Although the sample proportion $\hat{p}$ that we calculate probably does not equal p, the average of all the different sample proportions that we could have calculated (from all the different possible samples) is equal to p)

(3) has standard deviation $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ (exactly correct if the sampled population is infinite, approximately correct if the sampled population is finite and much larger than (say, at least 20 times as large as) the sample size)

where p is the population proportion for the category.

How to obtain the formulas?

- Population proportion is p.
- A sample has n elements, $X_1, X_2, \ldots, X_{n-1}, X_n$.
- The value of X_i may be 1 or 0. The probabilities that $X_i = 1$ is p and $X_i = 0$ is $(1-p)$.
- The mean of X_i is $\bar{X}_i = 1 \times p + 0 \times (1-p) = p$. (also known in Chapter 5)
- The variance of X_i is $\sigma_{X_i}^2 = p(1-\bar{X}_i)^2 + (1-p)(0-\bar{X}_i)^2 = p(1-p)$.
- The sample proportion is $\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$.
- The mean of $\hat{p}$ is $\mu_{\hat{p}} = \frac{1}{n}\sum_{i=1}^{n} \bar{X}_i = \frac{np}{n} = p$.
- The variance of $\hat{p}$ is $\sigma_{\hat{p}}^2 = \left(\frac{1}{n}\right)^2 \sum_{i=1}^{n} \sigma_{X_i}^2 = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$

Alternatively

- The variance of $\hat{p}$ is $\sigma_{\hat{p}}^2 = \left(\frac{1}{n}\sum_{i=1}^{n} X_i - \frac{1}{n}\sum_{i=1}^{n} \overline{X}_i\right)^2$

$$= \frac{\left(\sum_{i=1}^{n} X_i - \sum_{i=1}^{n} \overline{X_i}\right)^2}{n^2} = \frac{(n \times p \times (1-p)^2 + (n \times (1-p) \times (0-p)^2}{n^2}$$

$$= \frac{p(1-p)^2 + (1-p)(0-p)^2}{n}$$

$$= \frac{p(1-p)}{n}$$

Example 7.5 The Cheese Spread Case: Improving Profitability

- A food processing company developed a new cheese spread spout (pronounced: spowt) which may save production cost. If only less than 10% of current purchasers do not accept the design, the company would adopt and use the new spout.
- 1,000 current purchasers are randomly selected and inquired, and 63 of them say they would stop buying the cheese spread if the new spout were used. So, the sample proportion $\hat{p} = 0.063$.
- if $p = 0.1$, what is the probability of observing a sample of size 1,000 with sample proportion $\hat{p} \leq 0.063$?

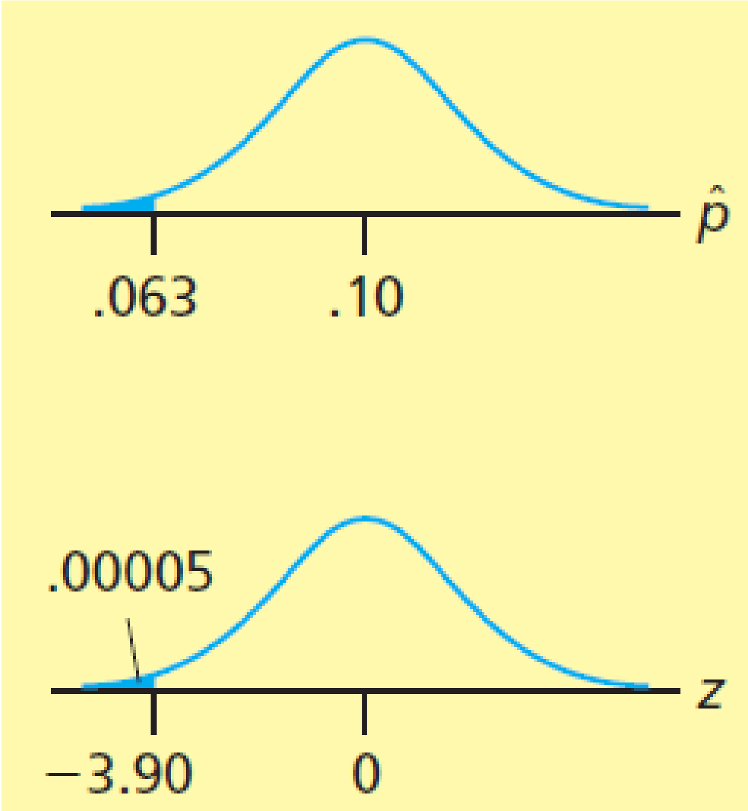
- If $p = 0.10$, since $n = 1{,}000$, $np \geq 5$ and $n(1-p) \geq 5$, $\hat{p}$ is approximately normal with

$$\mu_{\hat{p}} = p = 0.1,$$

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = 0.094868,$$

$$P(\hat{p} \leq 0.063 \text{ if } p = 0.1) = P\left(z \leq \frac{0.063 - \mu_{\hat{p}}}{\sigma_{\hat{p}}}\right)$$

$$= P\left(z \leq \frac{0.063 - 0.10}{0.094868}\right) = P(z \leq -3.90) \leq 0.00005.$$



- So, if $p = 0.1$, the chance of observing at most 63 out of 1,000 randomly selected customers do not accept the new design is less than 0.00005.
- But such observation does occur. This means that we have extremely strong evidence that $p \neq 0.1$, and p is in fact less than 0.1.
- In other words, we have extremely strong evidence that fewer than 10 percent of current purchasers would stop buying the cheese spread if the new spout were used.
- It seems that introducing the new spout will be profitable. Therefore, the company can adopt the new design.

Exercises for Section 7.3 (if time permits)

7.22 If the sample size n is large, the sampling distribution of $\hat{p}$ is approximately a normal distribution. What condition must be satisfied to guarantee that n is large enough to say that $\hat{p}$ is approximately normally distributed?

Answer: $np \geq 5$ and $n(1-p) \geq 5$

7.23 Describe the effect of increasing the sample size on the population of all possible sample proportions.

Answer: $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$, increasing n will reduce $\sigma_{\hat{p}}$

7.24 In each of the following cases, determine whether the sample size n is large enough to say that the sampling distribution of $\hat{p}$ is a normal distribution. (a) $p = 0.4$, $n = 100$. (b) $p = 0.1$, $n = 20$. (c) $p = 0.99$, $n = 400$.

Answer:

(a) $np = 40$ and $n(1-p) = 60$. Large enough

(b) $np = 2$ and $n(1-p) = 18$. Not large enough

(c) $np = 396$ and $n(1-p) = 4$. Not large enough

7.25 In each of the following cases, find the mean, variance, and standard deviation of the sampling distribution of the sample proportion $\hat{p}$.

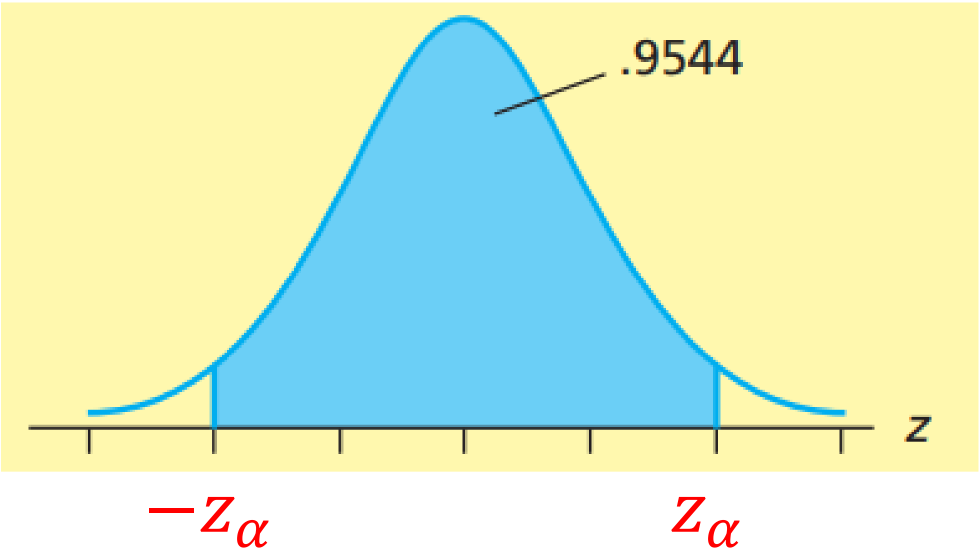
(a) $p = 0.5$, $n = 250$. (b) $p = 0.1$, $n = 100$.

Answer:

(a) $np = 125 \geq 5, n(1-p) = 125 \geq 5.$ So can use $\mu_{\hat{p}} = p = 0.5$, and $\sigma_{\hat{p}}^2 = \frac{p(1-p)}{n} = 0.001, \sigma_{\hat{p}} = \sqrt{\sigma_{\hat{p}}^2} = 0.0316.$

(b) $np = 10 \geq 5, n(1-p) = 90 \geq 5.$ So can use $\mu_{\hat{p}} = p = 0.1$, and $\sigma_{\hat{p}}^2 = \frac{p(1-p)}{n} = 0.0009, \sigma_{\hat{p}} = \sqrt{\sigma_{\hat{p}}^2} = 0.03.$

7.26 For each situation in Exercise 7.25, find an interval that contains approximately 95.44 percent of all the possible sample proportions.



Answer: for (a) only

Step 1: find $\mu_{\hat{p}} = 0.5, \sigma_{\hat{p}} = 0.03162.$

Step 2: Find z_α (a point) so that

$$P(-z_\alpha \leq z \leq z_\alpha) = 95.44.$$

$$P(z \leq -z_\alpha) = \frac{1-0.9544}{2} = 0.0228$$

From the table (or by calculation), $z_\alpha = 2.$

Step 3: Find interval for $\hat{p}$

$$z = \frac{\hat{p} - \mu_{\hat{p}}}{\sigma_{\hat{p}}}$$

$$\hat{p} = z\sigma_{\hat{p}} + \mu_{\hat{p}}$$

So: $0.4368 \leq \hat{p} \leq 0.5632$

7.27 Suppose that we will randomly select a sample of $n = 100$ elements from a population and that we will compute the sample proportion $\hat{p}$ of these elements that fall into a category of interest. If the true population proportion p equals 0.9:

(a) Describe the shape of the sampling distribution of $\hat{p}$. Why can we validly describe the shape?

(b) Find the mean and the standard deviation of the sampling distribution of $\hat{p}$.

Answer:

(a) It is approximately normal distributed. Since $np = 90 > 5$ and $n(1 - p) = 10 > 5$.

(b) $\mu_{\hat{p}} = p = 0.9$, $\sigma_{\hat{p}}^2 = \frac{p(1-p)}{n} = \frac{(0.9)(1-0.9)}{100} = 0.0009$, $\sigma_{\hat{p}} = \sqrt{\sigma_{\hat{p}}^2} = 0.03$.

7.30 On February 8, 2002, the Gallup Organization released the results of a poll concerning American attitudes toward the 19th Winter Olympic Games in Salt Lake City, Utah. The poll results were based on telephone interviews with a randomly selected national sample of 1,011 adults, 18 years and older, conducted February 4–6, 2002.

(a) Suppose we wish to use the poll's results to justify the claim that more than 30 percent of Americans (18 years or older) say that figure skating is their favorite Winter Olympic event. The poll actually found that 32 percent of respondents reported that figure skating was their favorite event. If, for the sake of argument, we assume that 30 percent of Americans (18 years or older) say figure skating is their favorite event (that is, $p = 0.3$), calculate the probability of observing a sample proportion of 0.32 or more; that is, calculate $P(\hat{p} \geq 0.32)$.

(b) Based on the probability you computed in part (a), would you conclude that more than 30 percent of Americans (18 years or older) say that figure skating is their favorite Winter Olympic event?

Solution

Checking the conditions

$p = 0.3, n = 1{,}011$

To find $P(\hat{p} \geq 0.32)$.

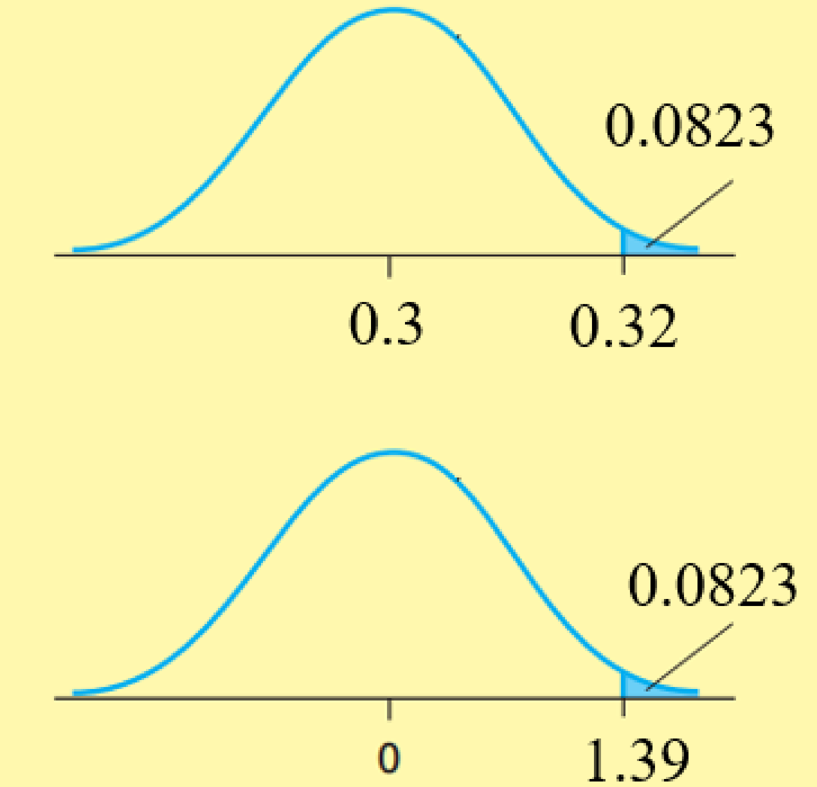
$\mu_{\hat{p}} = p = 0.3,$

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = 0.0144$$

$$z = \frac{0.32 - 0.3}{0.0144} = 1.39$$

Finding $P(\hat{p} \geq 0.32)$ is equivalent to finding $P(z \geq 1.39)$ or $P(z \leq -1.39)$.

$P(\hat{p} \geq 0.32 \text{ if } p = 0.3) = P(z \leq -1.39) = 0.0823.$



Hard to conclude that more than 30 percent of Americans say that figure skating is their favorite Winter Olympic event.

7.31 *Quality Progress* reports on improvements in customer satisfaction and loyalty made by Bank of America. A key measure of customer satisfaction is the response (on a scale from 1 to 10) to the question: “Considering all the business you do with Bank of America, what is your overall satisfaction with Bank of America?” Here, a response of 9 or 10 represents “customer delight”.

(a) Historically, the percentage of Bank of America customers expressing customer delight has been 48%. Suppose that we wish to use the results of a survey of 350 Bank of America customers to justify the claim that more than 48% of all current Bank of America customers would express customer delight. The survey finds that 189 of 350 randomly selected Bank of America customers express customer delight. If, for the sake of argument, we assume that the proportion of customer delight is $p = 0.48$, calculate the probability of observing a sample proportion greater than or equal to 189 / 350 = 0.54. That is, calculate $P(\hat{p} \geq 0.54)$.

(b) Based on the probability you computed in part (a), would you conclude that more than 48 percent of current Bank of America customers express customer delight? Explain.

Solution

Checking the conditions

$p = 0.48, n = 350$

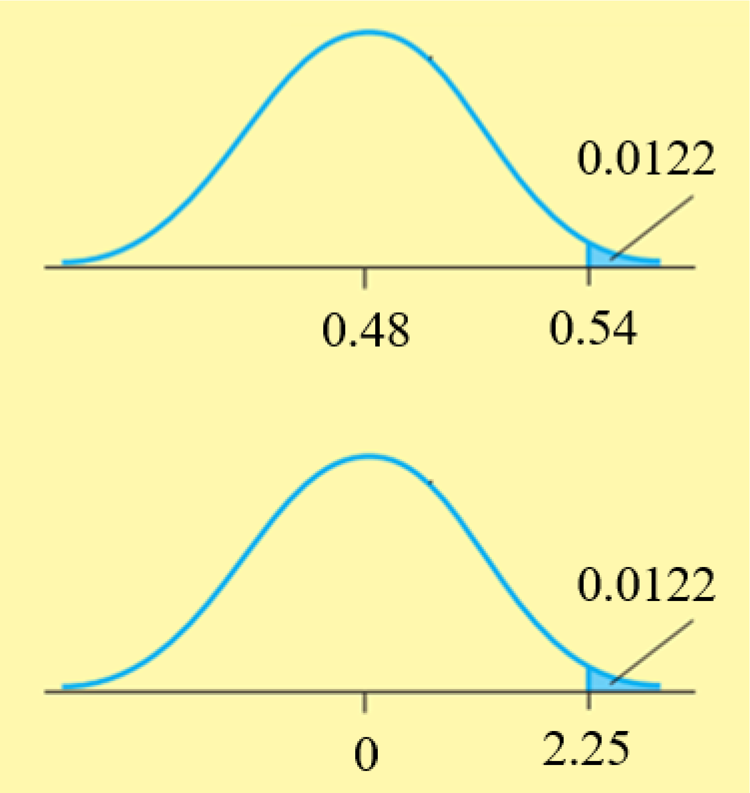
$\mu_{\hat{p}} = p = 0.48,$

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = 0.0267$$

$$z = \frac{0.54-0.48}{0.0267} = 2.25$$

Finding $P(\hat{p} \geq 0.54)$ is equivalent to finding $P(z \geq 2.25)$ or $P(z \leq -2.25)$.

$P(\hat{p} \geq 0.54 \text{ if } p = 0.48) = P(z \leq -2.25) = 0.0122$



The probability (chance) is small, but it happens.

So, conclude that more than 48 percent of current Bank of America customers express customer delight.

7.32 Again consider the survey of 350 Bank of America customers discussed in Exercise 7.31, and assume that 48% of Bank of America customers would currently express customer delight. That is, assume $p = 0.48$. Find.

(a) The probability that the sample proportion obtained from the sample of 350 Bank of America customers would be within three percentage points of the population proportion. That is, find $P(0.45 \leq \hat{p} \leq 0.51)$.

(b) The probability that the sample proportion obtained from the sample of 350 Bank of America customers would be within six percentage points of the population proportion. That is, find $P(0.42 \leq \hat{p} \leq 0.54)$.

Solution

Checking the conditions

$$p = 0.48, n = 350$$

$$\mu_{\hat{p}} = p = 0.48,$$

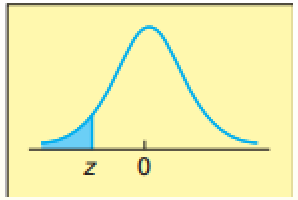
$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = 0.0267$$

$$z = \frac{\hat{p} - \mu_{\hat{p}}}{\sigma_{\hat{p}}}$$

(a) Finding $P(0.45 \le \hat{p} \le 0.51)$ is equivalent to finding $P(-1.12 \le z \le$

From the table, $P(0.45 \le \hat{p} \le 0.51) = 1 - 2 \times 0.1314 = 0.7372$

TABLE A.3 **Cumulative Areas under the Standard Normal Curve**

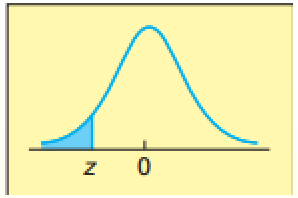


z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379

(b) Finding $P(0.42 \leq \hat{p} \leq 0.54)$ is equivalent to finding $P(-2.25 \leq \hat{p} \leq$

From the table, $P(0.42 \leq \hat{p} \leq 0.54) = 1 - 2 \times 0.0122 = 0.9756$

TABLE A.3 Cumulative Areas under the Standard Normal Curve



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379

7.33 Based on your results in Exercise 7.32, would it be reasonable to state that the survey's "margin of error" is 3 percentage points? 6 percentage points? Explain.

Answer

(a) No, From the table, $P(0.45 \leq \hat{p} \leq 0.51) = 0.7372$. This probability is not reasonably high. So the margin of error" is 3 percentage points is not reasonable.

(a) **Yes**, From the table, $P(0.42 \leq \hat{p} \leq 0.54) = 0.9756$. This probability is reasonably high. So the margin of error" is 6 percentage points is reasonable.

Thank you!